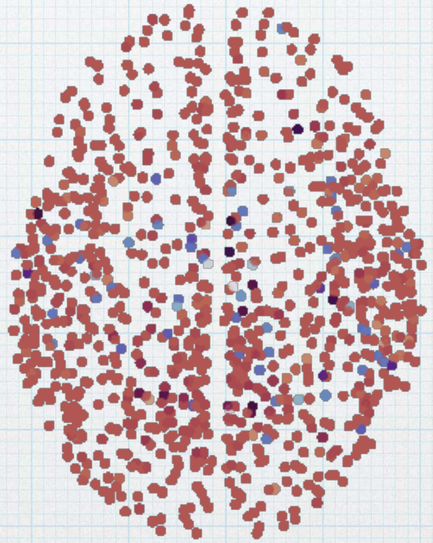
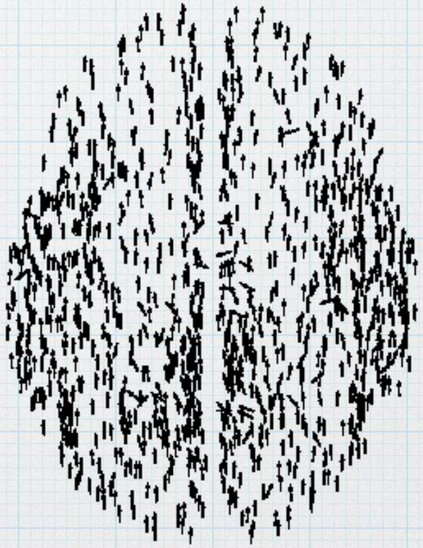
Hodge theory on graphs:
An application to biophysics and to the brain

Ramón Nartallo-Kaluarachchi
Research Associate, DAMTP
University of Cambridge

with David Beers, UCLA





Functions and flows on graphs

Some systems are naturally modelled over *continuous spaces* and some over *discrete spaces*, like graphs or simplicial complexes

We are going to generalise some basic calculus operations from continuous spaces to discrete:

Lim (2020) SIAM Review

Functions and flows on graphs

Some systems are naturally modelled over *continuous spaces* and some over *discrete spaces*, like graphs or simplicial complexes

We are going to generalise some basic calculus operations from continuous spaces to discrete:

$$G = (V, E)$$

V is the set of vertices $\{1,2,...,n\}$

E is the set of edges $\{(i,j)\}$, where we assign each edge an orientation e.g. (i,j) if $i < j$

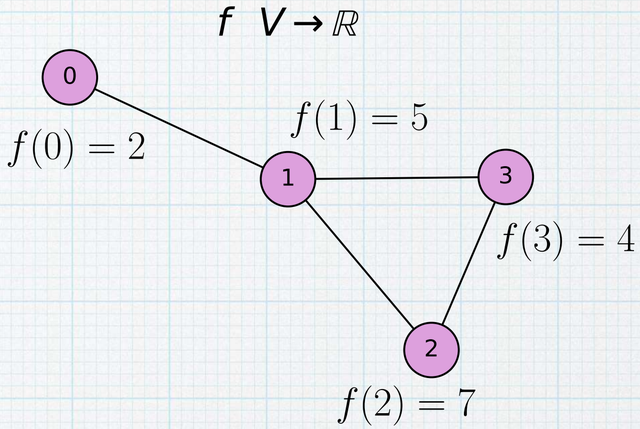
Lim (2020) SIAM Review

Functions and flows on graphs

We will start by defining a real-valued function on the vertices

$$f : V \to \mathbb{R}$$

e.g. $f(2) = 5$



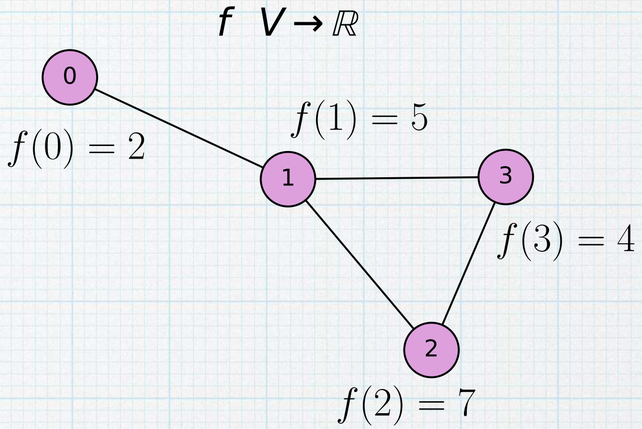
Lim (2020) SIAM Review

Functions and flows on graphs

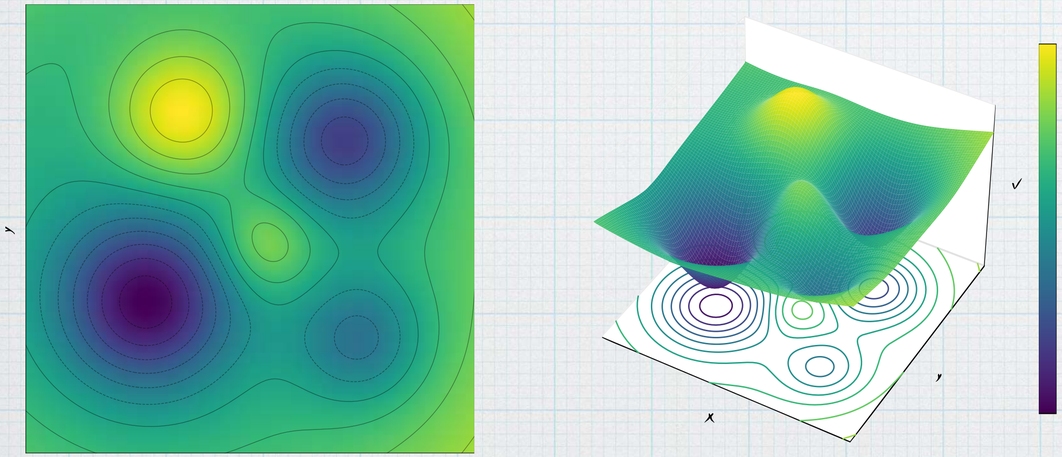
We will start by defining a real-valued function on the vertices

$$f : V \to \mathbb{R}$$

e.g. $f(2) = 5$



$$u(x, y)$$

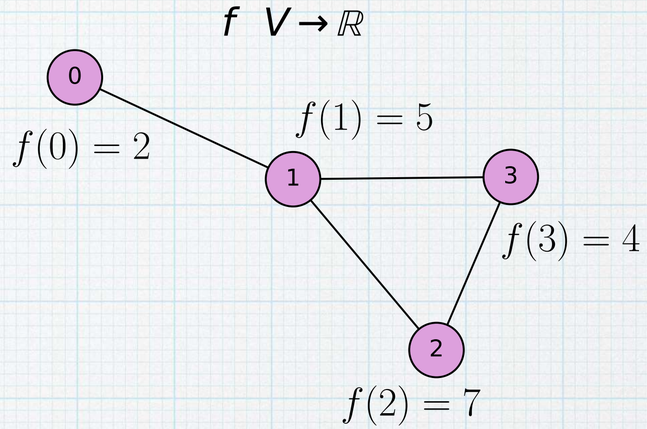


Functions and flows on graphs

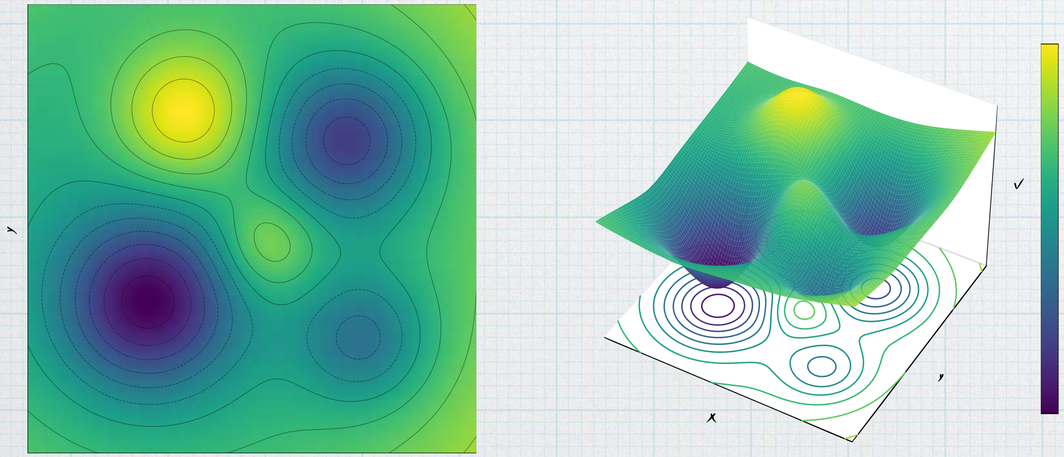
We will start by defining a real-valued function on the vertices

$$f : V \to \mathbb{R}$$

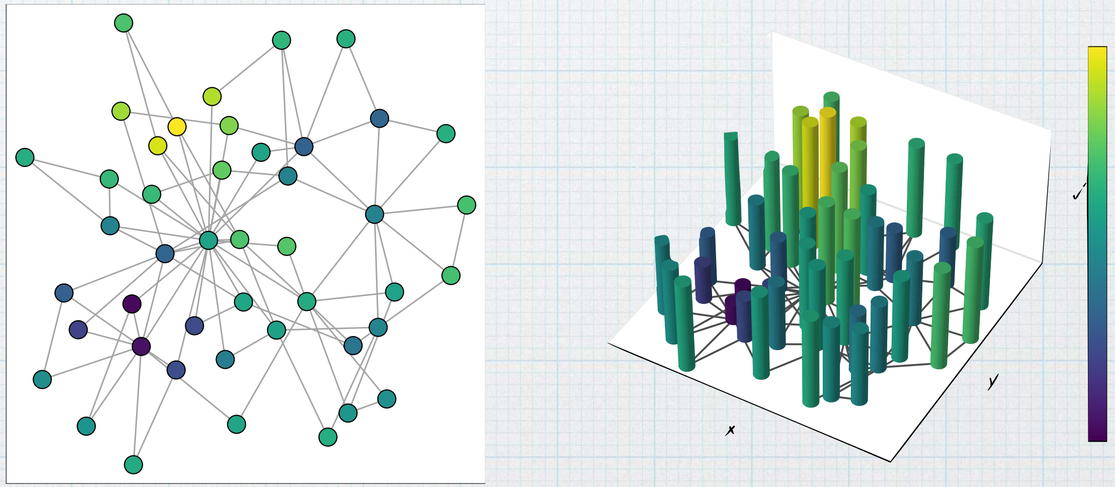
e.g. $f(2) = 5$



$u(x, y)$



$f(i)$



Lim (2020) SIAM Review

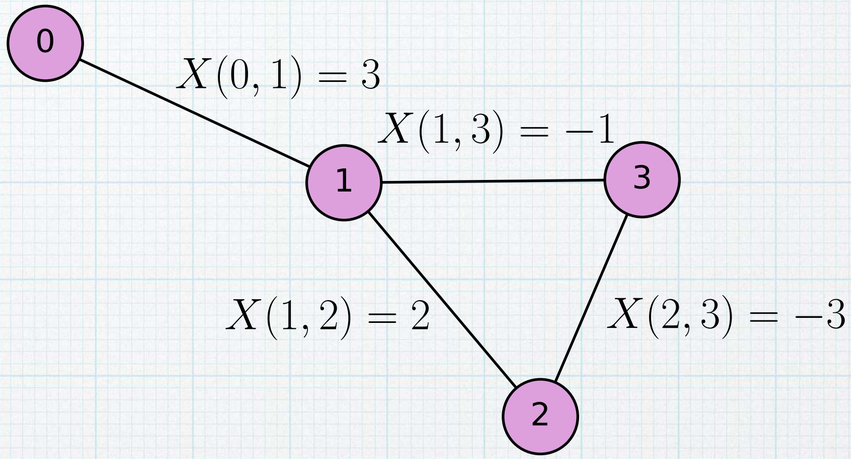
Functions and flows on graphs

Next, we have *flows* on edges

$$X : E \to \mathbb{R}$$

Our flow must be *alternating*

$$X(i,j) = -X(j,i)$$



It is zero between nodes that are not connected i.e. $X(i,j) = 0$ if $(i,j) \notin E$

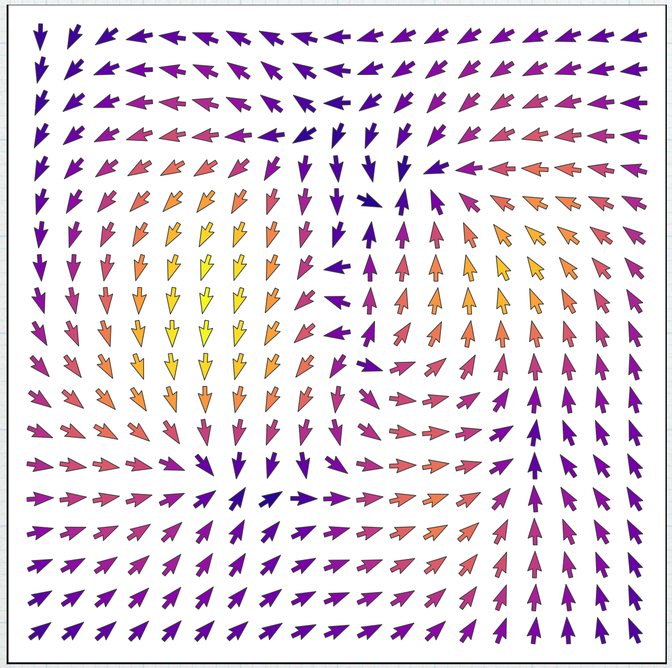
Lim (2020) SIAM Review

Functions and flows on graphs

Next, we have *flows* on edges

$$X : E \to \mathbb{R} \qquad X(i,j) = -X(j,i)$$

$$\mathbf{v}(x,y)$$



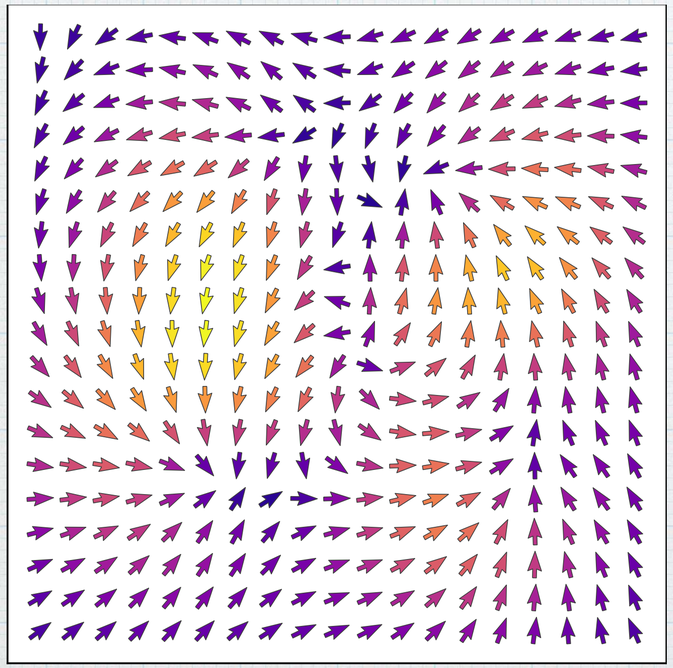
Lim (2020) SIAM Review

Functions and flows on graphs

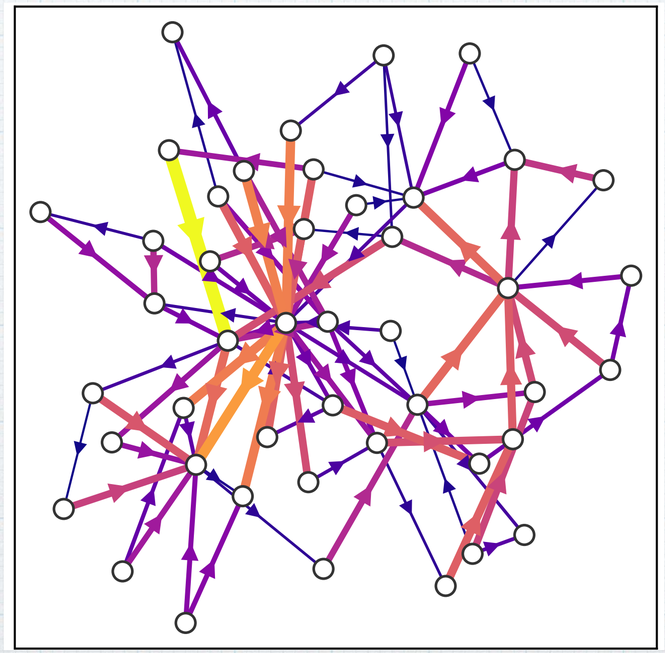
Next, we have *flows* on edges

$$X : E \to \mathbb{R} \qquad X(i,j) = -X(j,i)$$

$\mathbf{v}(x,y)$



$X(i,j)$



Lim (2020) SIAM Review

Discrete operators on graphs

With flows and functions, we can start to define discrete operators on graphs.

Let $H(V)$ be the space of all the functions on the vertices

Let $H(E)$ be the space of all the flows on edges

Note that these spaces can be identified with real spaces and be represented by vectors:

$H(V) \cong \mathbb{R}^{|V|}$

$H(E) \cong \mathbb{R}^{|E|}$

Discrete operators on graphs

With flows and functions, we can start to define discrete operators on graphs.

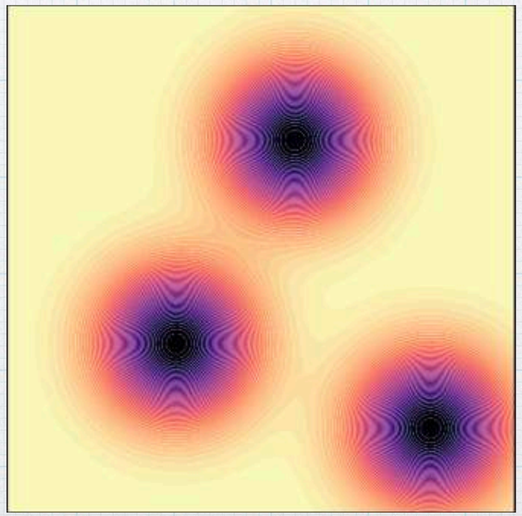
Let $H(V)$ be the space of all the functions on the vertices

Let $H(E)$ be the space of all the flows on edges

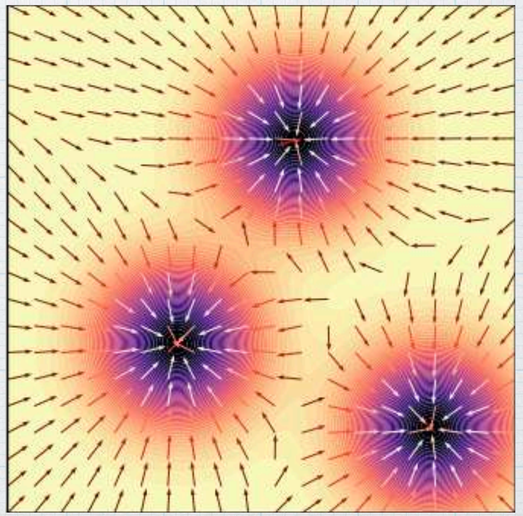
We have the **gradient operator**, grad : $H(V) \to H(E)$

$\text{grad}(f)(i,j) = f(j) - f(i)$

$u(x,y)$



$-\nabla u(x,y)$



Discrete operators on graphs

With flows and functions, we can start to define discrete operators on graphs.

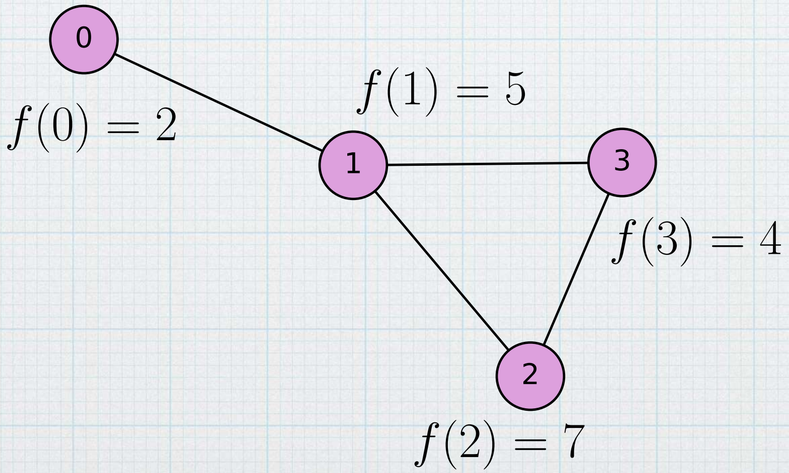
Let $H(V)$ be the space of all the functions on the vertices

Let $H(E)$ be the space of all the flows on edges

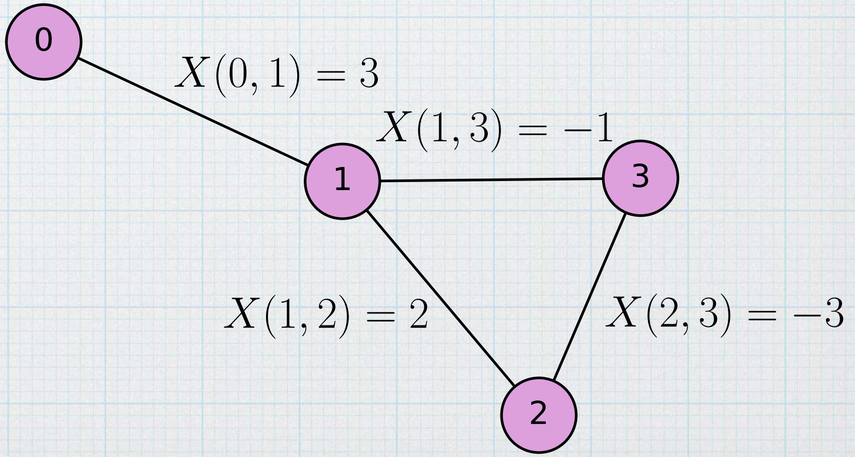
We have the **gradient operator**, grad : $H(V) \to H(E)$

$\text{grad}(f)(i,j) = f(j) - f(i)$

$f(i)$

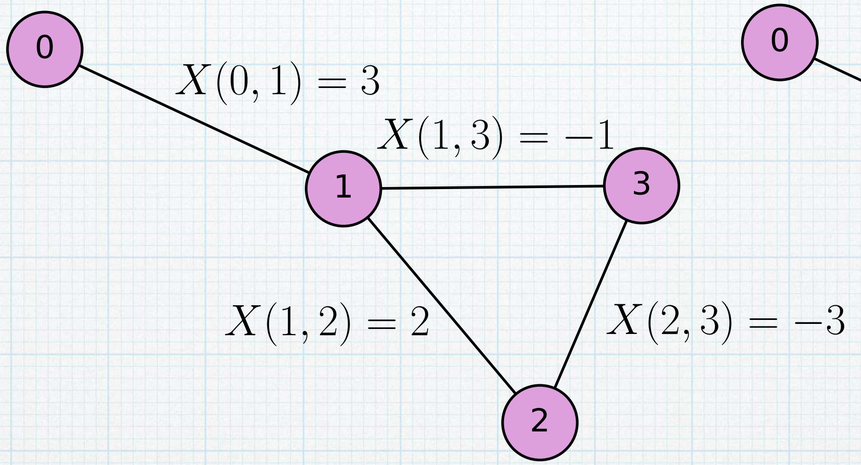


$\text{grad}(f)(i,j)$

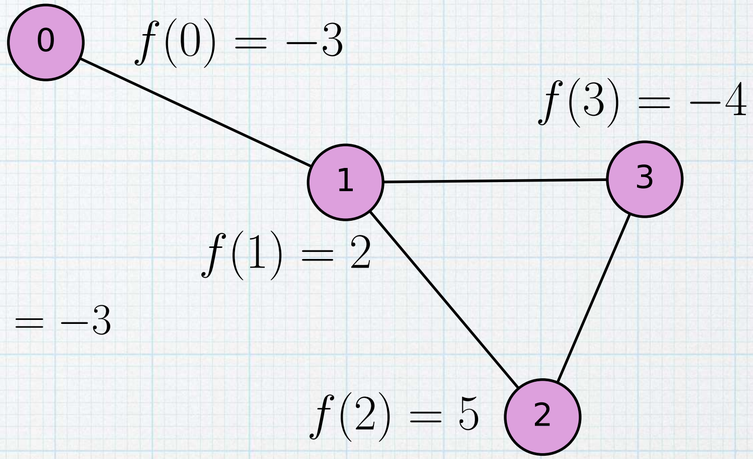


We have the **divergence operator**, div : $H(E) \to H(V)$

$$\operatorname{div}(X)(i) = \sum_{j=1}^{N} X(i,j)$$

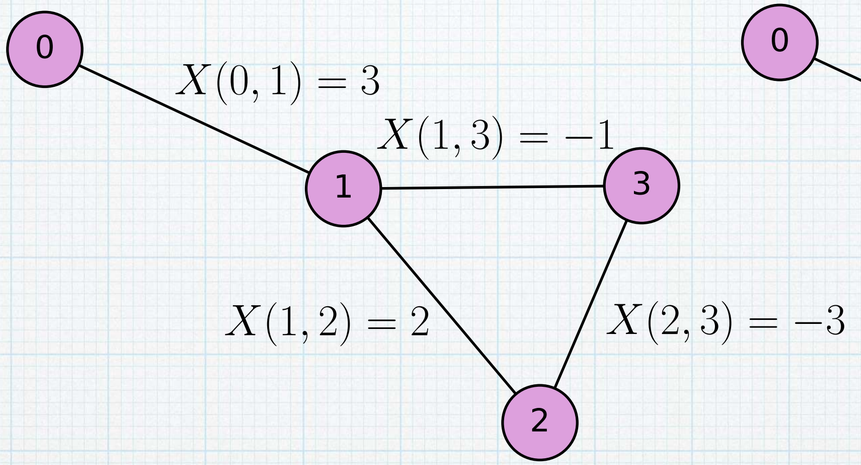


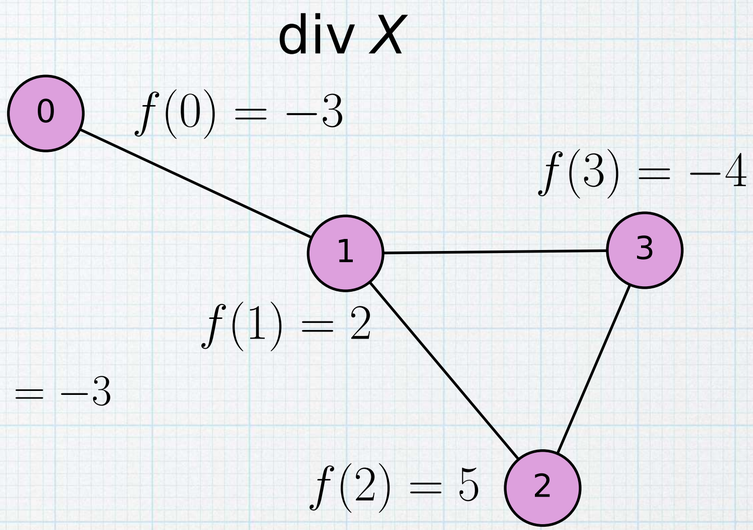
div X



We have the **divergence operator**, div : $H(E) \to H(V)$

$$\text{div}(X)(i) = \sum_{j=1}^{N} X(i,j)$$





As flows are real-valued vectors, operators can be represented as real-valued matrices

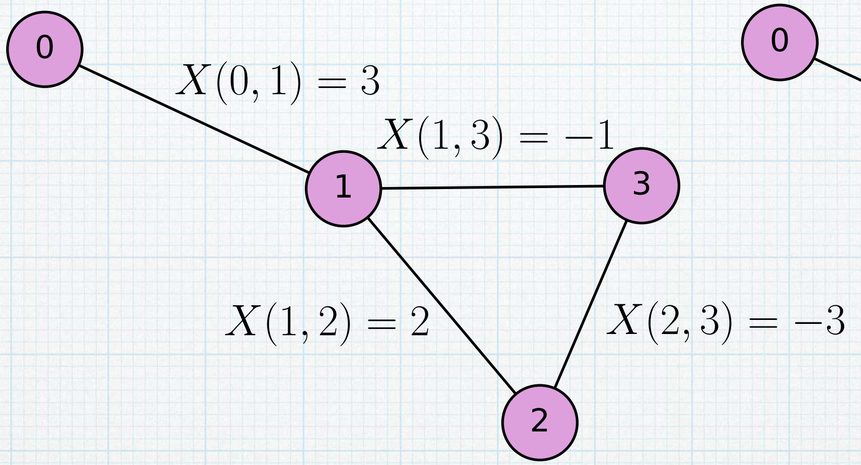
grad $\cong B_1^\top$

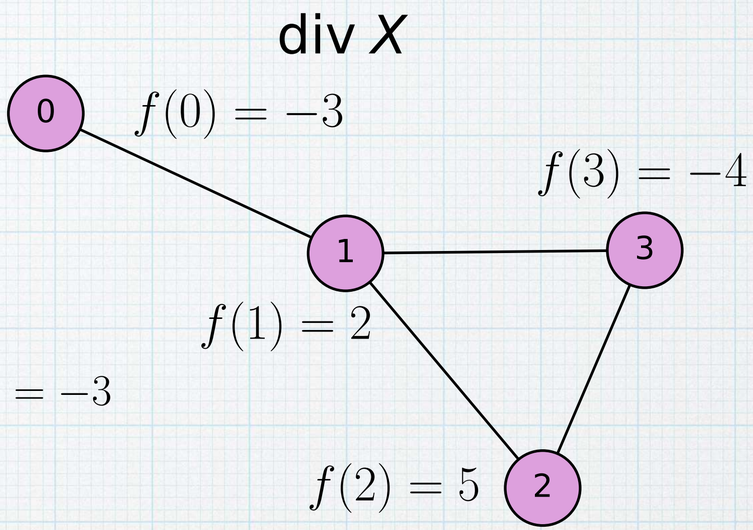
div $\cong -B_1$

Where B_1 is the incidence matrix

We have the **divergence operator**, div : $H(E) \to H(V)$

$$\text{div}(X)(i) = \sum_{j=1}^{N} X(i,j)$$





Example:

The continuous Laplacian is $\Delta = -\nabla \cdot \nabla$

The discrete Laplacian is $\Delta_0 = -$ divgrad

$$(\Delta_0 f)(i) = \sum_{j:(i,j)\in E} f(i) - f(j) = k_i f(i) - \sum_{j:(i,j)\in E} f(j) = k_i f(i) - \sum_{j=1}^{N} A_{ij} f(j)$$

As a matrix, the Laplacian is $L = D - A$; the combinatorial graph Laplacian

Different inner products

We can **reweigh** flows by defining an inner product on the spaces of functions on vertices and edges

$$\langle f, g \rangle_V = \sum_i w_i f(i) g(i)$$

$$\langle X, Y \rangle_E = \sum_{i<j} w_{ij} X(i,j) Y(i,j)$$

The divergence operator is different

$$\text{div}(X)(i) = \sum_{j=1}^{N} \frac{w_{ij}}{w_i} X(i,j)$$

This means that the matrices for the divergence and the Laplacian are different now.

Different inner products

Example:

Let's reweigh nodes by *degree* $w_i = k_i$

$$(\Delta_0 f)(i) = f(i) - \sum_{j=1}^{N} \frac{A_{ij}}{k_i} f(j)$$

As a matrix this becomes,

$$\hat{L}_0 = I - D^{-1}A$$

This is the **random walk Laplacian**— not symmetric but can be symmetrised

$$\tilde{L}_0 = D^{1/2}\hat{L}_0 D^{-1/2} = D^{-1/2}LD^{-1/2}$$

This is the **normalised Laplacian**

Flows on triangles

We can also define flows around triangles, $\phi(i,j,k)$

The space of these is $H(T)$ which has inner product

$$\langle \phi, \psi \rangle = \sum_{i<j<k} w_{ijk} \phi(i,j,k) \psi(i,j,k)$$

We can define the curl and its adjoint (which again generalises the continuous version)

$$(\text{curl } X)(i,j,k) = X(i,j) + X(j,k) + X(k,i)$$

$$(\text{curl}^* \phi)(i,j) = \sum_k \frac{w_{ijk}}{w_{ij}} \phi(i,j,k)$$

$$\phi(i,j,k) = \phi(j,k,i) = \phi(k,i,j) = -\phi(j,i,k) = -\phi(i,k,j) = -\phi(k,j,i),$$

Lim (2020) SIAM Review

Hodge Laplacian

The last ingredient we need is the **Hodge Laplacian**, which generalises the vector Laplacian, and gives us the **Hodge decomposition**

$\Delta_1 = -\,\text{grad div} + \text{curl}^*\text{curl}$; $\Delta_1 : H(E) \to H(E)$

Which has matrix form (using standard IP)

$$L_1 = B_1^\top B_1 + B_2 B_2^\top$$

Lim (2020) SIAM Review

Hodge Laplacian

The space of flows on edges decomposes:

$$X(i,j) = (\text{grad}\, f)(i,j) + R(i,j)$$

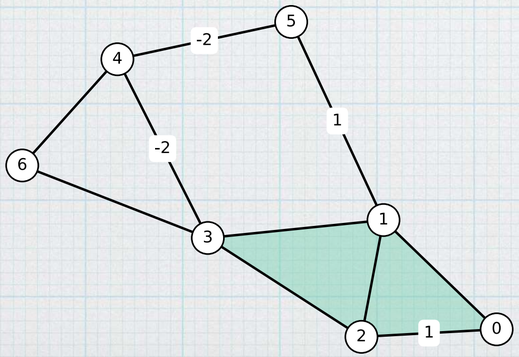
Where $R(i,j) \in \ker(\text{div})$ i.e. it is divergence free

We can also split it further

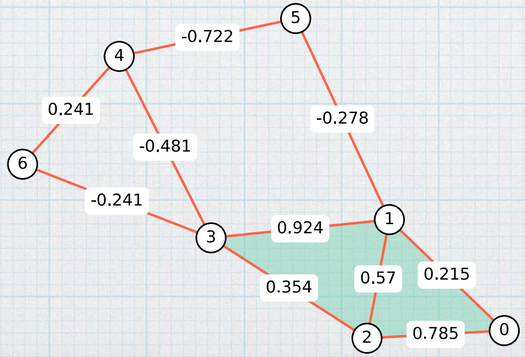
$$X(i,j) = (\text{grad}\, f)(i,j) + (\text{curl}^* \phi)(i,j) + X_H(i,j),$$

Where $X_H \in \ker(\Delta_1)$

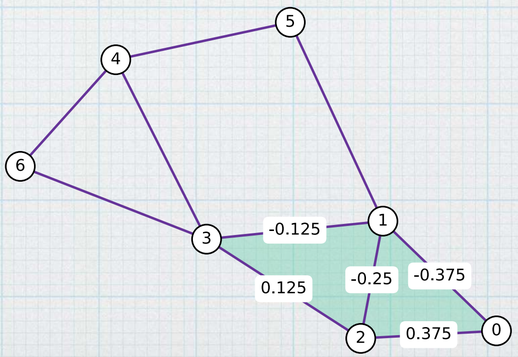
$X = \text{grad}\, f + \text{curl}^* \phi + X_H$



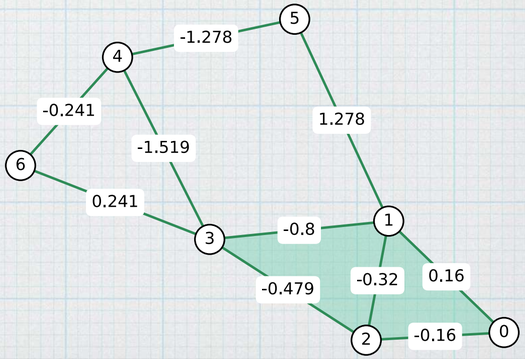
grad f



curl* ϕ



X_H



Hodge Laplacian

Because the operators are just sparse matrices and the spaces are **orthogonal**, solving the Hodge decomposition is just a least-squares problem

$$X(i,j) = (\text{grad}\, f)(i,j) + R(i,j)$$

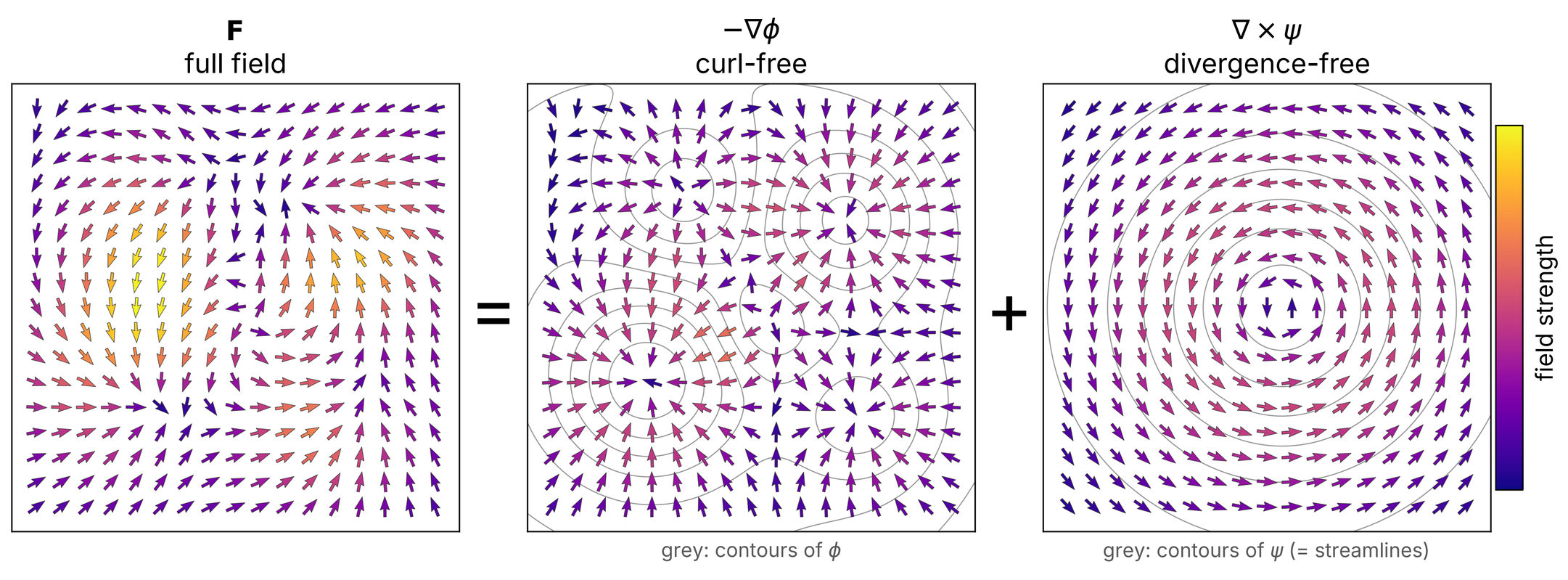
$$f = \text{argmin}_g \,||\,\text{grad}\, g - X\,||^2$$

$$X(i,j) = (\text{grad}\, f)(i,j) + (\text{curl}^*\phi)(i,j) + X_H(i,j),$$

$$f = \text{argmin}_g \,||\,\text{grad}\, g - X\,||^2$$
$$\phi = \text{argmin}_\psi \,||\,\text{curl}^*\psi - X\,||^2$$

The Helmholtz decomposition

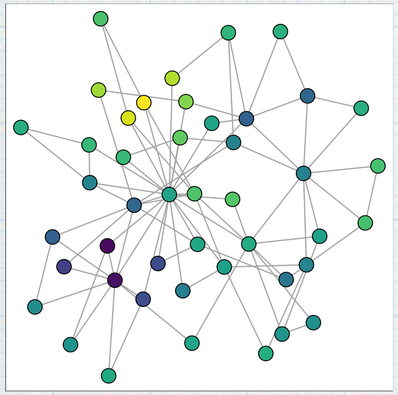
It is reminiscent of the **Helmholtz decomposition** from vector calculus and electrodynamics

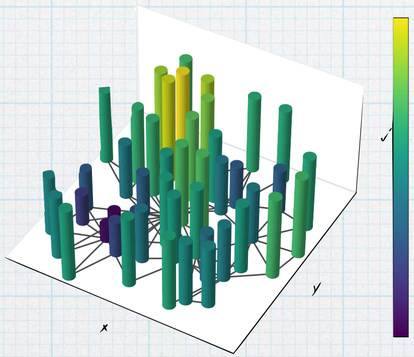


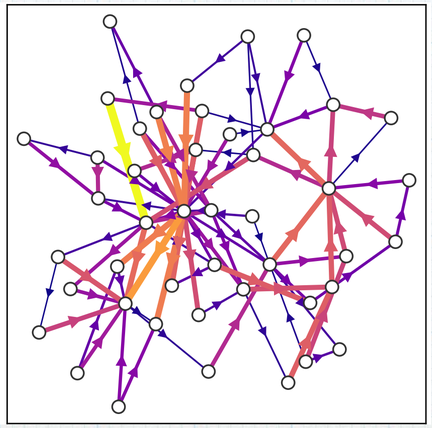
Applications

Ranking in directed/weighted networks/tournaments

In a ‘perfect’ competition, participants would beat each other by precisely the differences in their ‘levels’ i.e. the **flow** would be a perfect gradient e.g. $X(i,j) = W_{ij} - W_{ji}$





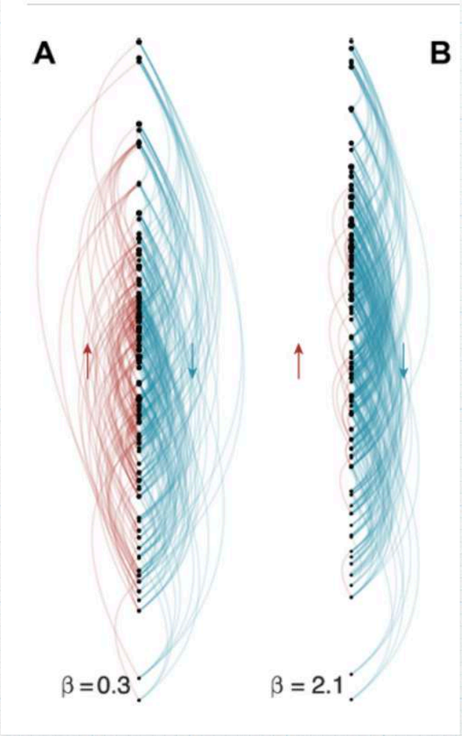


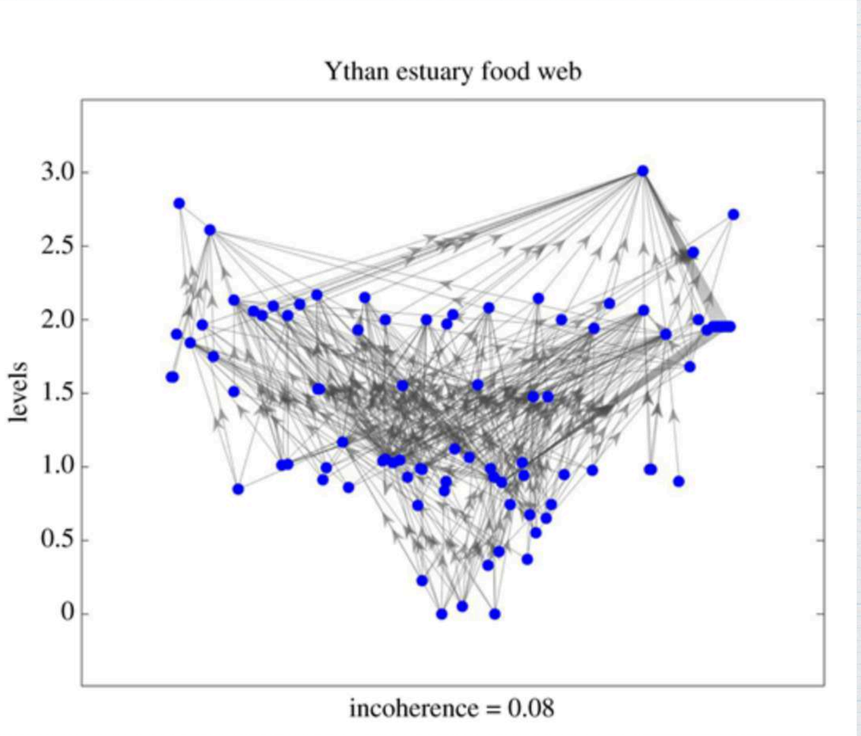
This can be used to extract the ‘cyclic’ component of competitions or to rank nodes by their ‘trophic’ level

Applications

- **‘Hodge rank’ —** Jiang et al. (2009) *Mathematical Programming*
- **‘Spring rank’—** De Bacco et al. (2018) *Science Advances*
- **‘Trophic coherence’—** Johnson et al. (2014) *PNAS,* MacKay et al. (2020) *RSOS*

Review: Strang et al. (2022) *SIAM Review*





Project 1:
Decomposing stochastic trajectories with the HHD

RNK, Expert, Beers, Strang, Kringelbach, Lambiotte, Goriely (2024) Decomposing force-fields as flows on graphs reconstructed from stochastic trajectories, *Proceedings of the Third Learning on Graphs conference (LoG).* ‘Best Paper’

Irreversible stochastic processes

Consider a stochastic differential equation (SDE)

$$dx(t) = f(x)\,dt + \sigma(x)\,dw(t)$$

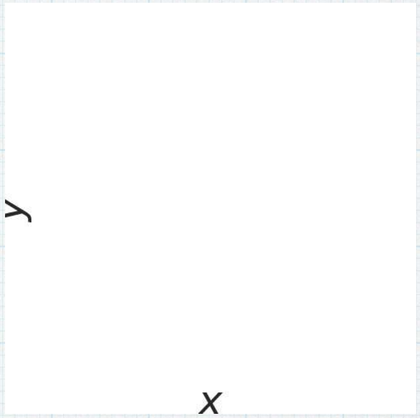
The density dynamics are given by the Fokker Planck equation

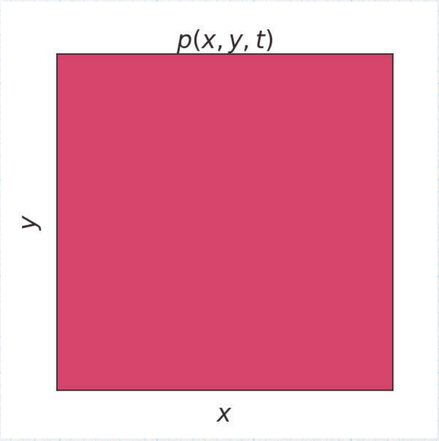
$$\partial_t p(x,t) = -\nabla \cdot J(x,t)$$

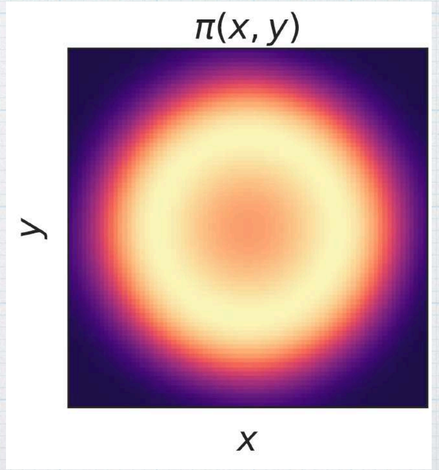
$$J(x,t) = f(x)p(x,t) - \nabla \cdot (D(x)p(x,t)), \text{ where } D(x) = \frac{1}{2}\sigma(x)\sigma^{\mathsf{T}}(x)$$

The process is stationary when the probability density no longer changes i.e.

$$\nabla \cdot J_{\mathrm{ss}} = 0,\ \pi(x)$$







RNK et al. (2026) *Physics Reports*

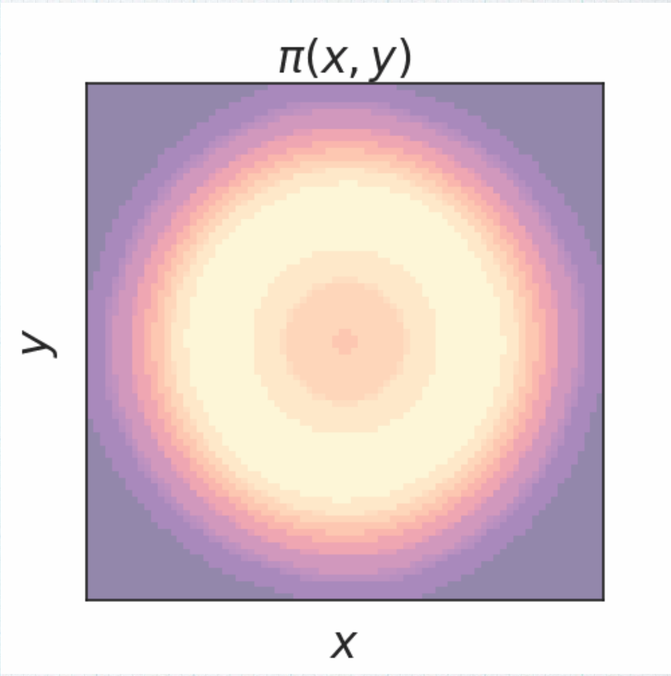
Irreversible stochastic processes

The process is in steady-state but it is not **still**

Here we can differentiate between **reversible** (equilibrium) and **irreversible** (nonequilibrium) steady-states

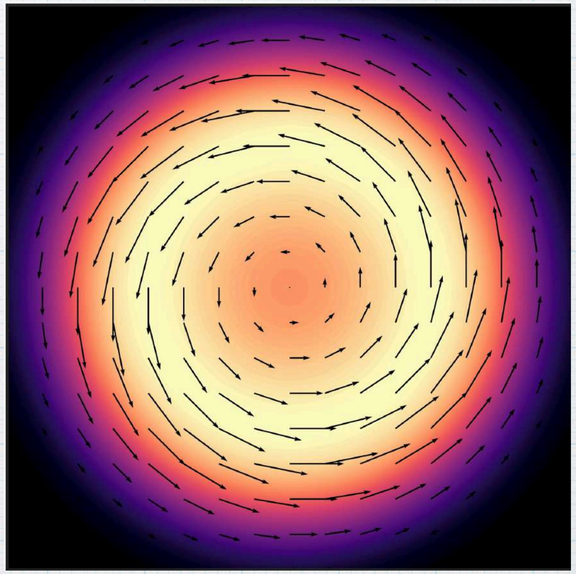
Reversible: $J_{\mathrm{ss}} = 0$

Irreversible: $J_{\mathrm{ss}} \neq 0$



There are very deep links here to nonequilibrium physics and the thermodynamic nature of living systems

Schrödinger (1944) *What is life?*
England (2020) *Every Life Is On Fire: How Thermodynamics Explains the Origins of Living Things*



RNK et al. (2026) *Physics Reports*

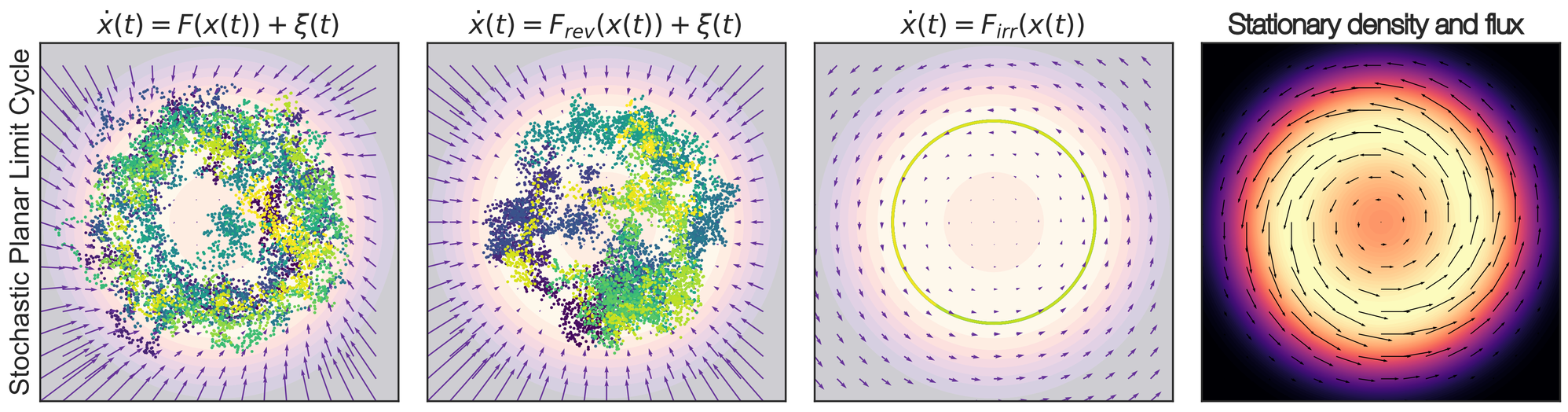
Irreversible stochastic processes

Stationary SDEs also admit a kind of Hodge decomposition (the HHD)

$f = f_{\text{rev}} + f_{\text{irr}}$

$f_{\text{rev}} = D\nabla \log \pi + \nabla \cdot D(x)$; this is **even** under time-reversal

$f_{\text{irr}} = J_{\text{ss}}/\pi$; this is **odd** under time-reversal



RNK et al. (2026) *Physics Reports*

Question: given stochastic trajectories from an SDE, can we decompose them into **irreversible** and **reversible** components using the Hodge decomposition

$$f = \epsilon \nabla \log \pi + J_{\mathrm{ss}}/\pi$$

$$X(i,j) = (\mathrm{grad}\, f)(i,j) + R(i,j)$$

RNK et al. (2024) *LoG*

Question: given stochastic trajectories from an SDE, can we decompose them into **irreversible** and **reversible** components using the Hodge decomposition

$$f = \epsilon \nabla \log \pi + J_{\mathrm{ss}}/\pi$$

$$X(i,j) = (\mathrm{grad}\, f)(i,j) + R(i,j)$$

1. Build a graph
2. Define a flow

RNK et al. (2024) *LoG*

Question: given stochastic trajectories from an SDE, can we decompose them into **irreversible** and **reversible** components using the Hodge decomposition

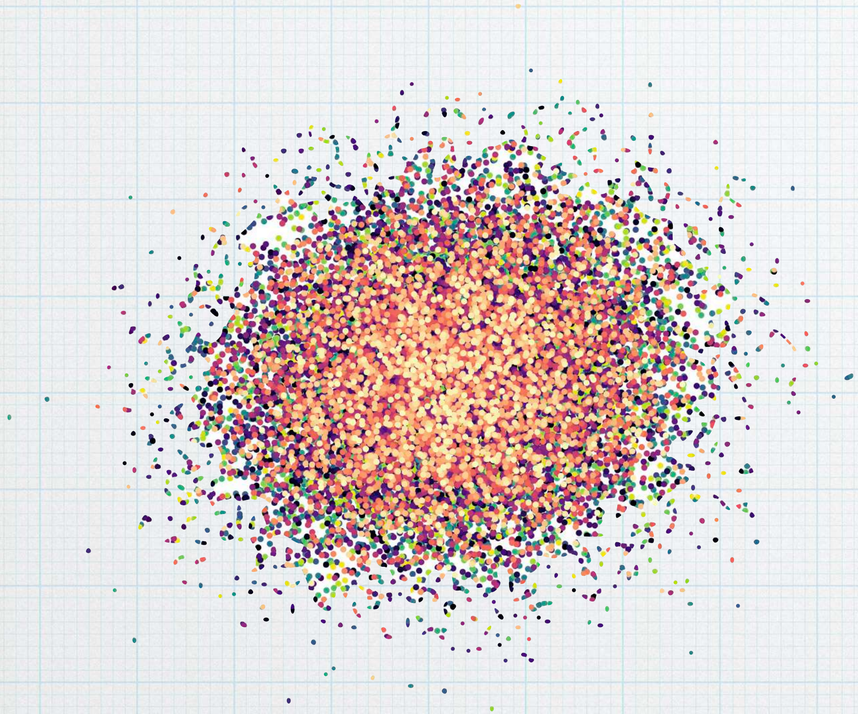
$$f = \epsilon \nabla \log \pi + J_{\mathrm{ss}}/\pi$$

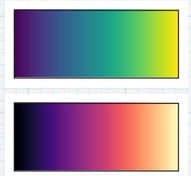
$$X(i,j) = (\mathrm{grad}\, f)(i,j) + R(i,j)$$

1. **Build a graph**
2. **Define a flow**

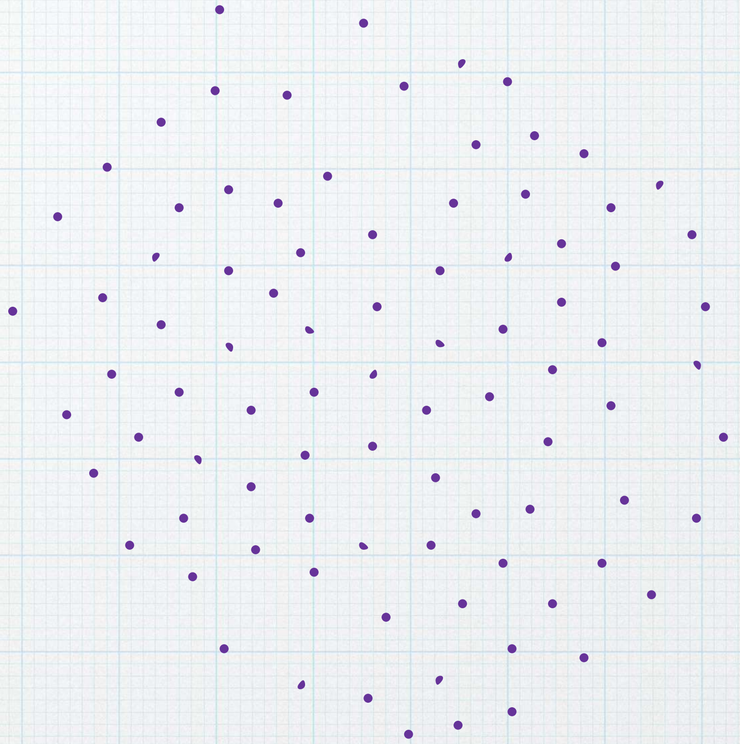
RNK et al. (2024) *LoG*

Building the graph





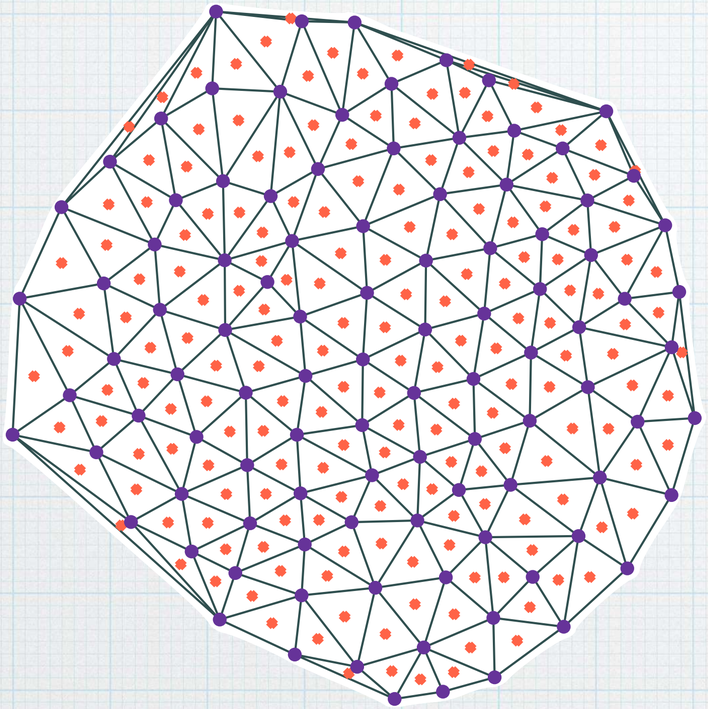
Step. 1 Begin with trajectories as a point-cloud in phase-space



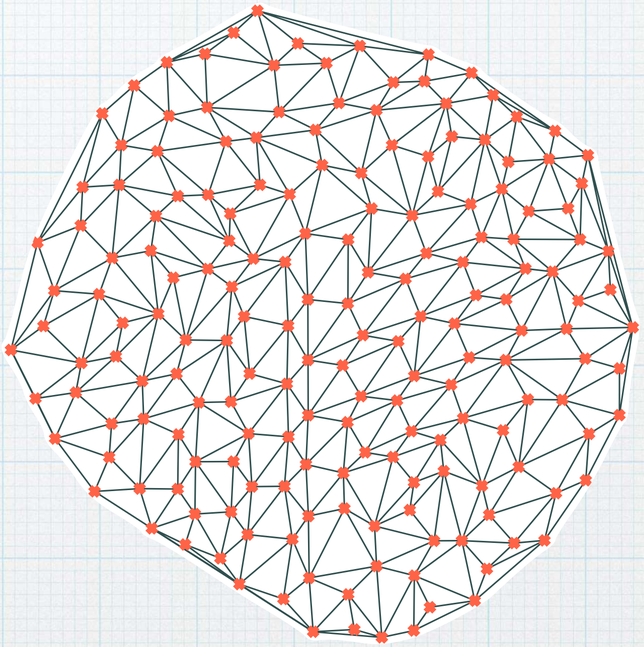
Step. 2 Perform farthest-point sub-sampling to get a uniform density sub-sample

RNK et al. (2024) *LoG*

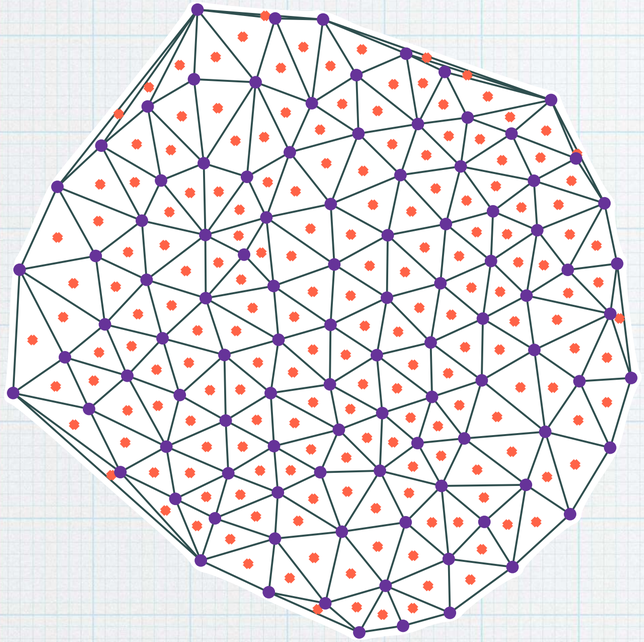
Building the graph

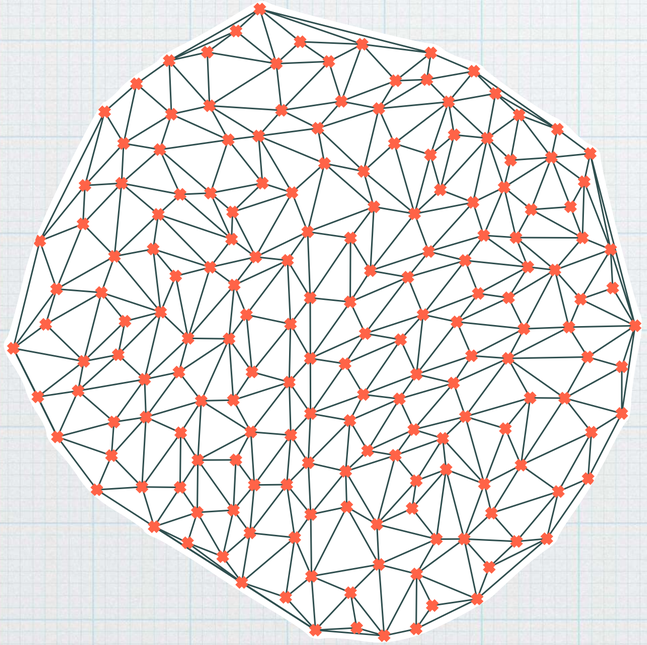


Step. 3 Triangulate space with Delaunay



Step. 4 Take its dual tessellation





What is the correct flow?

$$X(i,j)$$

The answer (courtesy of Alex Strang)

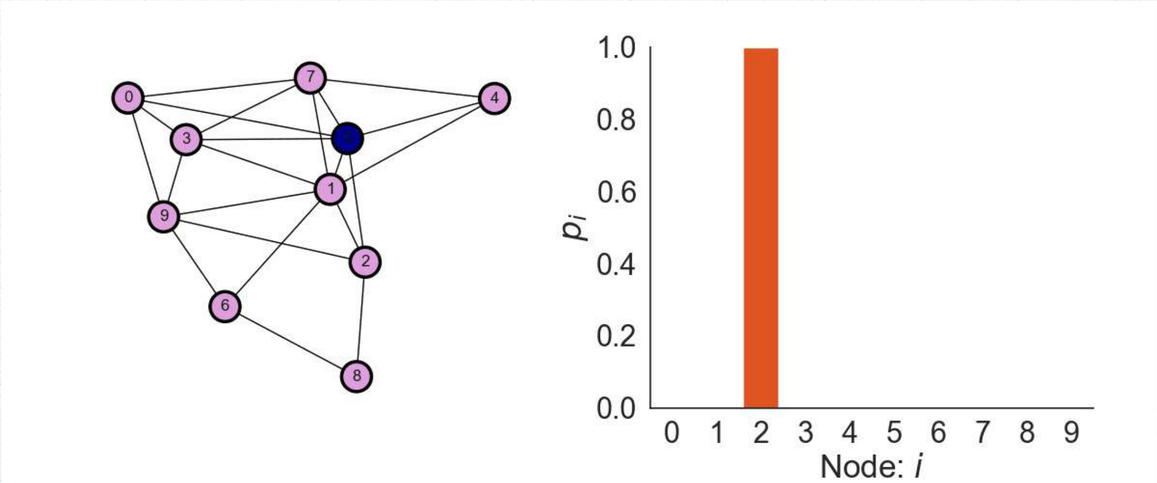
We can also define **discrete-state** irreversible stochastic processes

$$\frac{dp}{dt} = Lp$$

$$L\pi = 0$$

Reversible steady-state if and only if

$L_{ij}\pi_j = L_{ji}\pi_i$ — the **detailed balance** condition



Strang (2020) *Case Western Reserve University*

The answer (courtesy of Alex Strang)

Theorem (Strang 2020)

Given a stationary continuous-time Markov chain with Laplacian L, the process is **reversible** if and only if

$$X(i,j) = \log \sqrt{\frac{L_{ij}}{L_{ji}}}$$

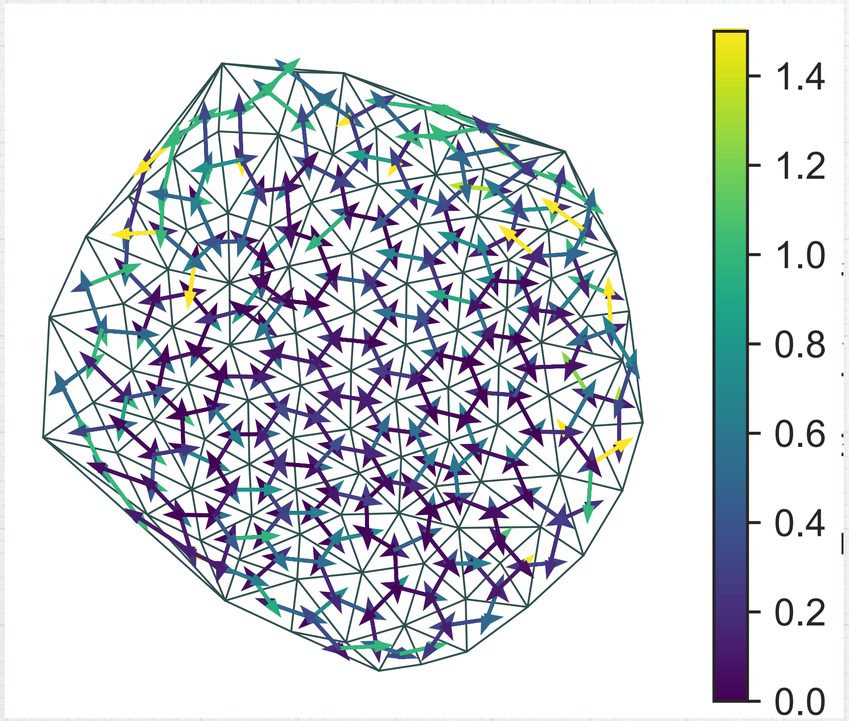
Is a gradient flow i.e. you can write $X = \text{grad}\, f$ for some $f \in H(V)$

This implies that if $X = \text{grad}\, f + R$, R is the **irreversible** component

$$X(i,j) = (\text{grad}\, f)(i,j) + R(i,j)$$
$$f = \epsilon \nabla \log \pi + J_{\text{ss}}/\pi$$

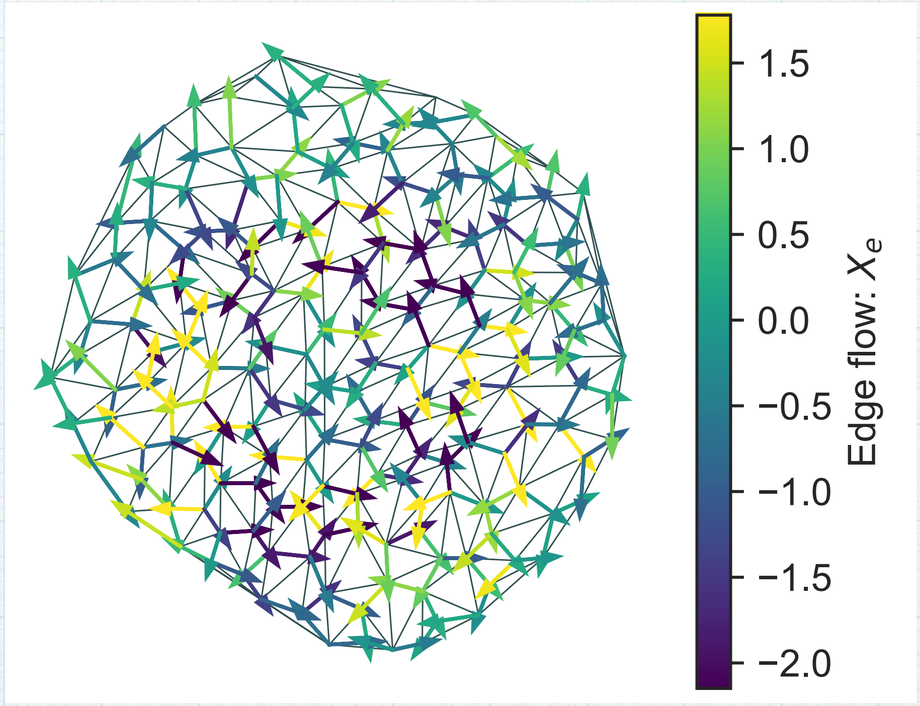
Strang (2020) *Case Western Reserve University*

$$\hat{L}_{ij} = \frac{N_{j \to i}}{T_j}$$



Step. 5 Infer the transition rates from the trajectories using MLE

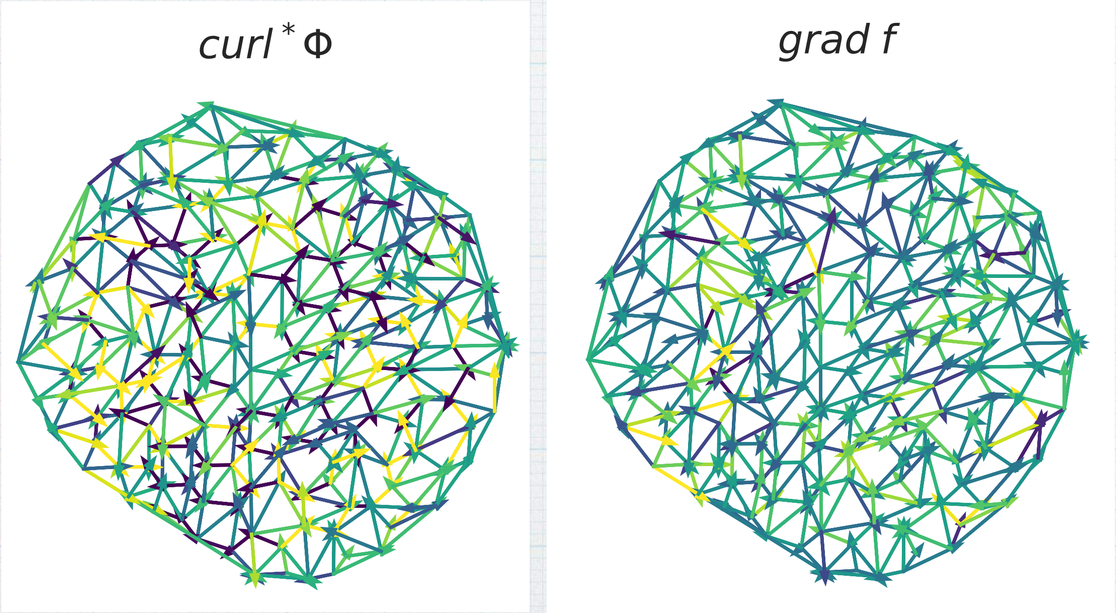
$$X(i,j) = \log \sqrt{\frac{L_{ij}}{L_{ji}}}$$



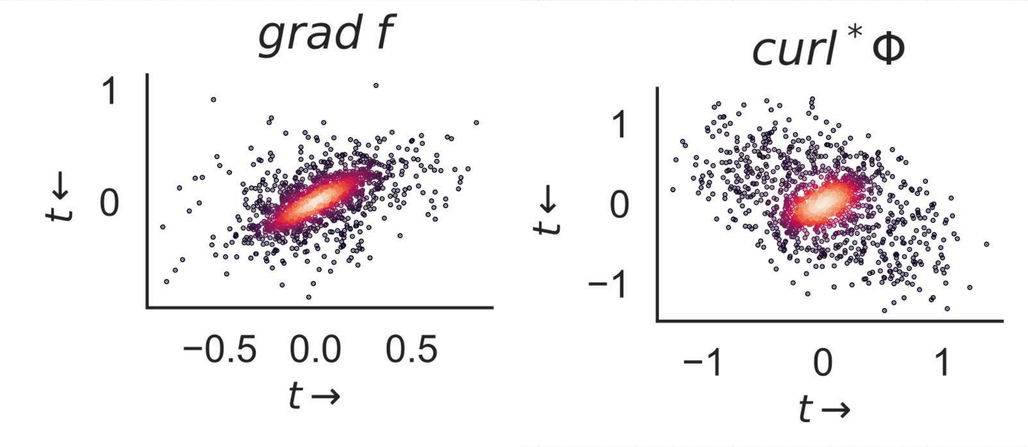
Step. 6 Define the edge flow

Step. 7 Define the edge-flow

$$X = \text{grad}\, f + R = \text{grad}\, f + \text{curl}^{*}\phi$$



Numerical experiments



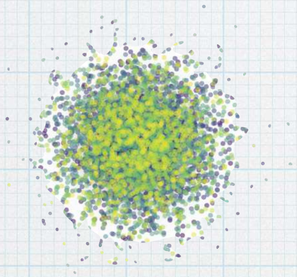
Our components are (approximately) odd and even under time-reversal respectively, as they are in the HHD

$$P_c = \frac{||X_c||}{||X_c|| + ||X_g||}$$

Now we will focus on the size of the irreversible component

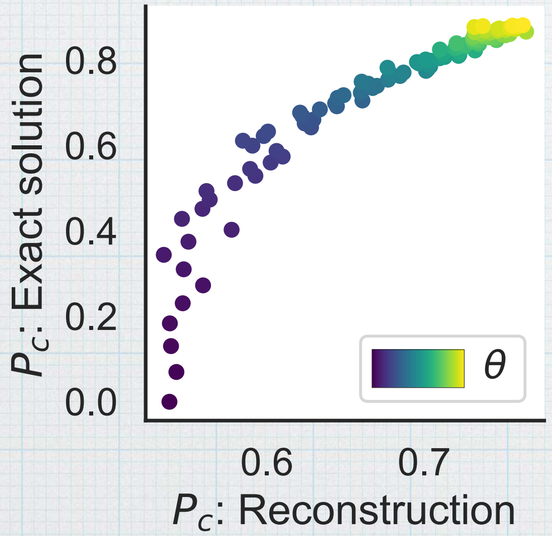
Numerical experiments

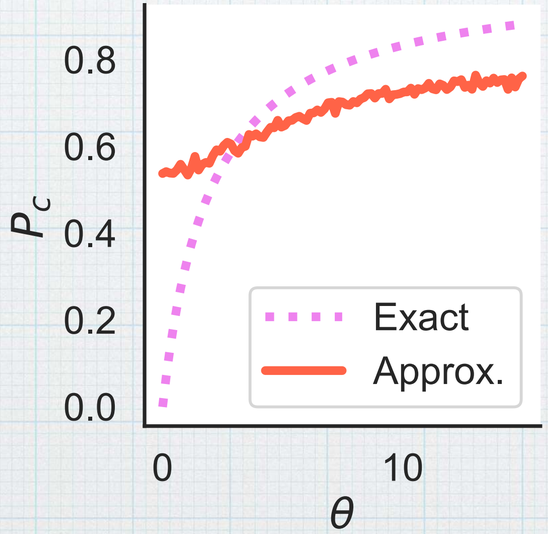
We consider two processes with solvable stationary distributions and decompositions

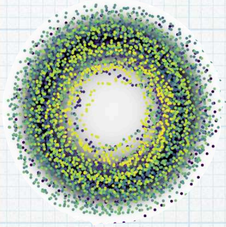


a)

OU: $\sigma = 1$

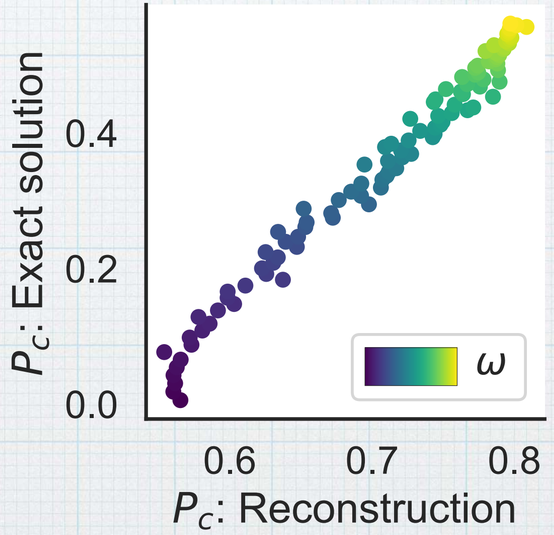


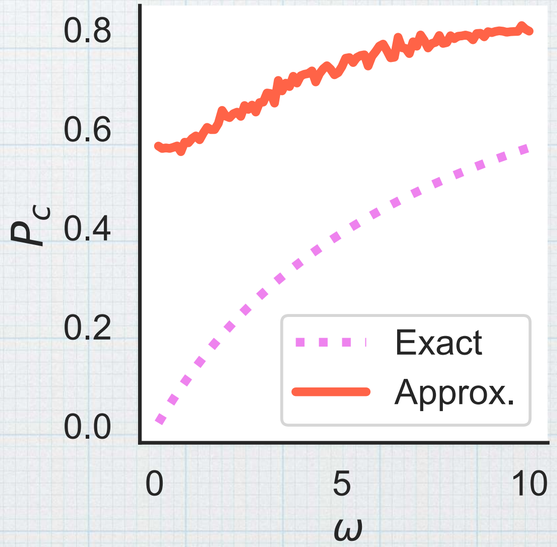




b)

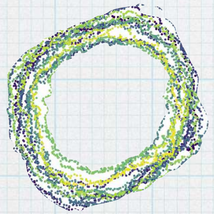
HO: $a = 1; \sigma = 1/\sqrt{2}$

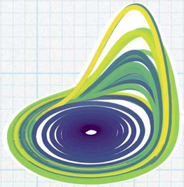




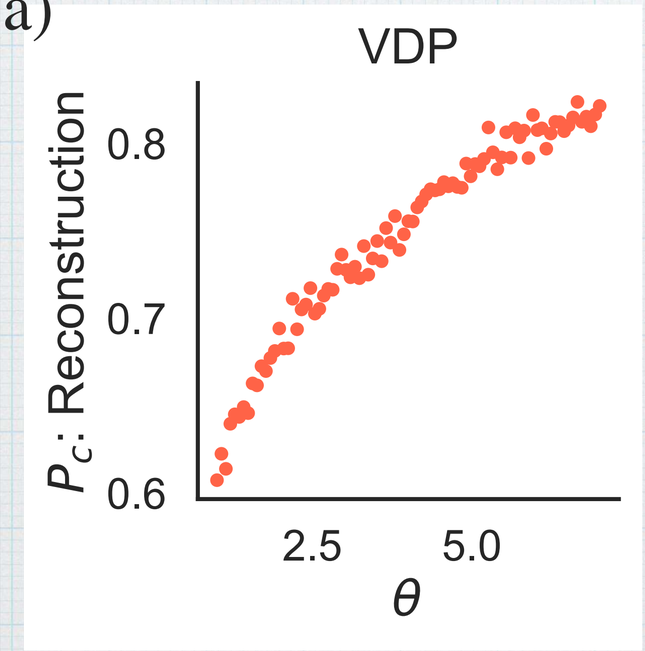
Numerical experiments

Next we consider two processes without solvable stationary distributions and decompositions

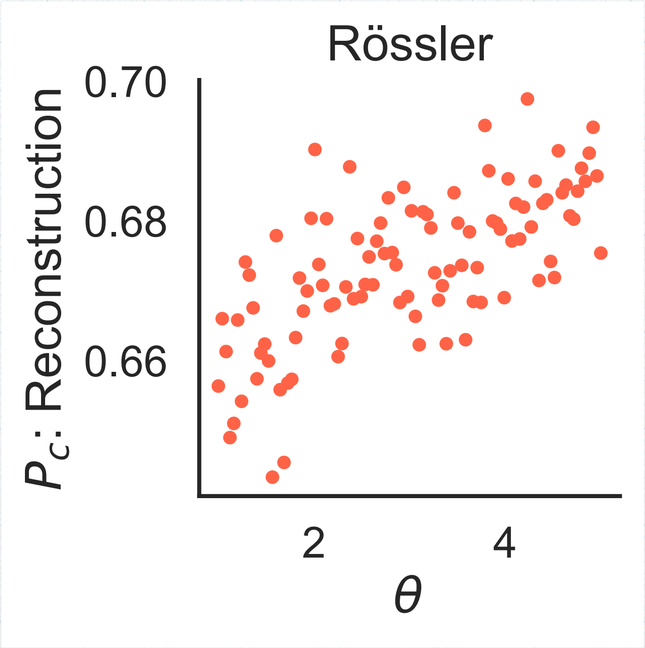




a)



b)



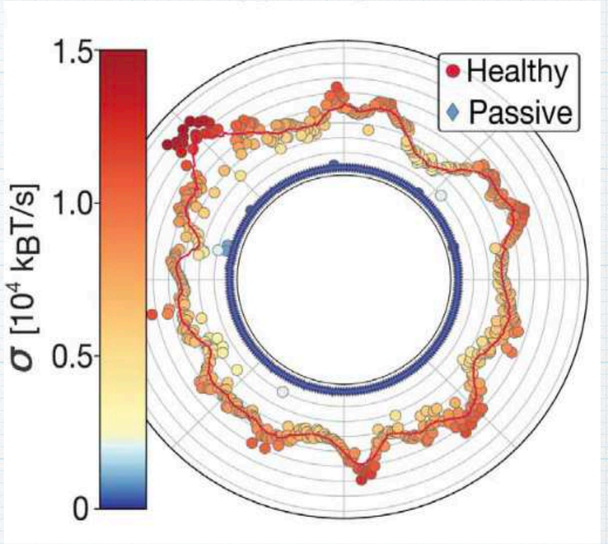
Di Terlizzi et al. (2024) *Science;* Costa et al. (2005) *PRL*

Real-world data analysis

Flickering red-blood cells:

ATP-depleted (passive) RBCs have lower dissipation than active/healthy counterparts

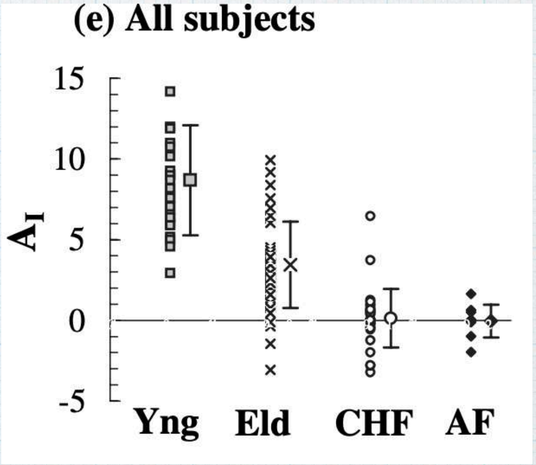
Data: outer-membrane position over time



Human heartbeats:

Arrhythmic heartbeats (myocardial infarction) have lower time-irreversibility than healthy counterparts

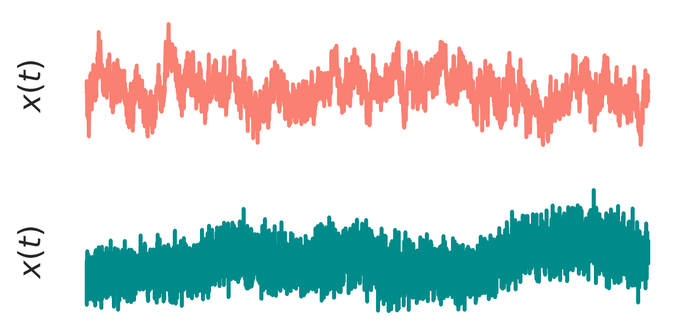
Data: electrocardiogram

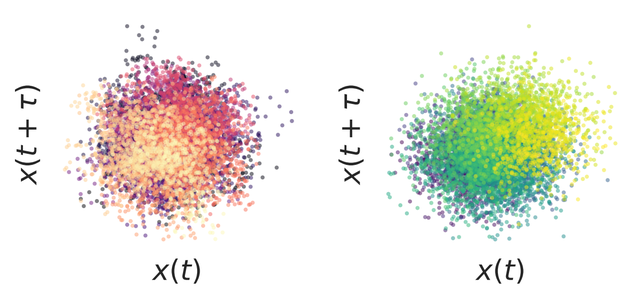


Real-world data analysis

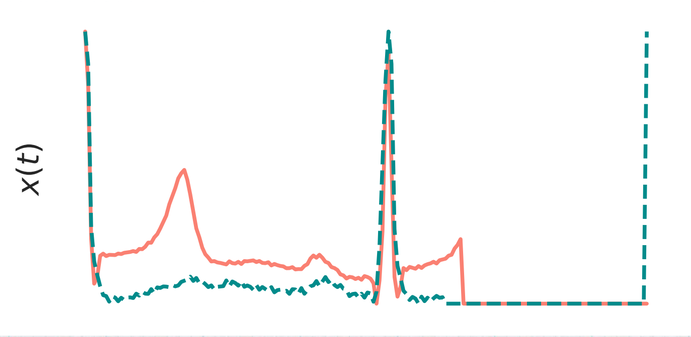
Reconstruct phase-dynamics using a time-delay embedding

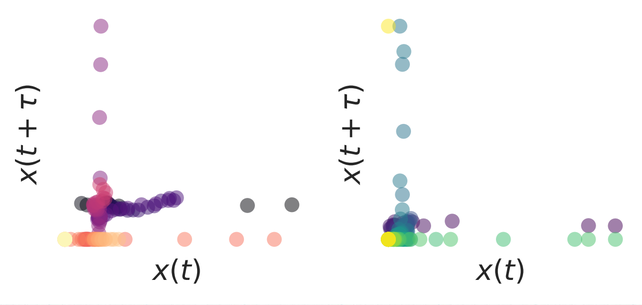
$$y(t) = (x(t), x(t+\tau))$$





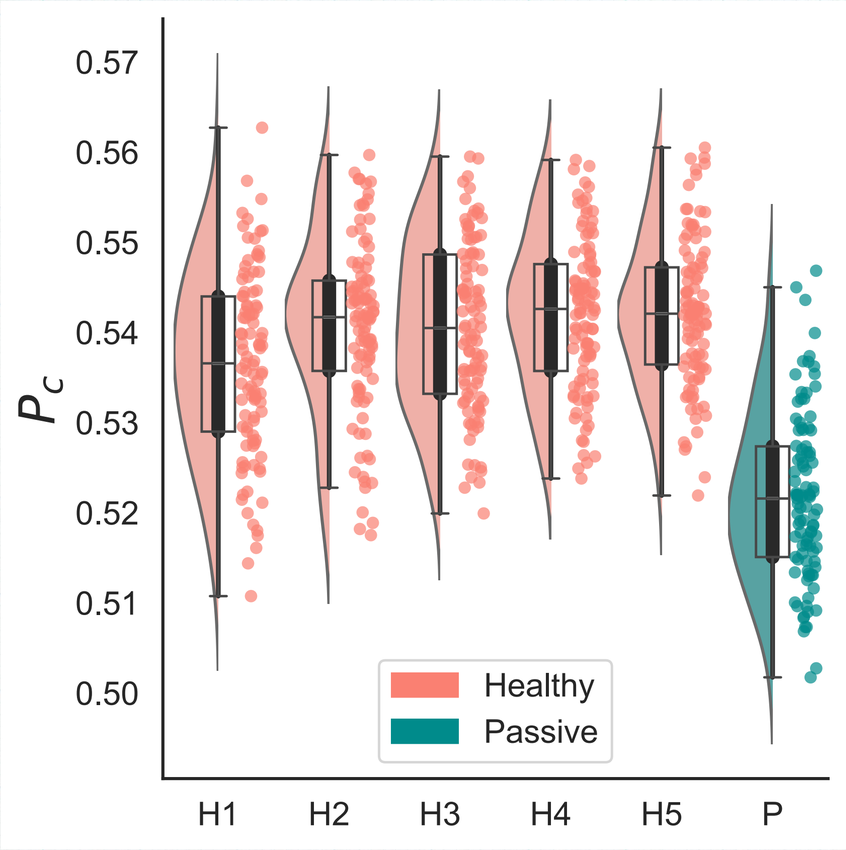
Healthy
Passive

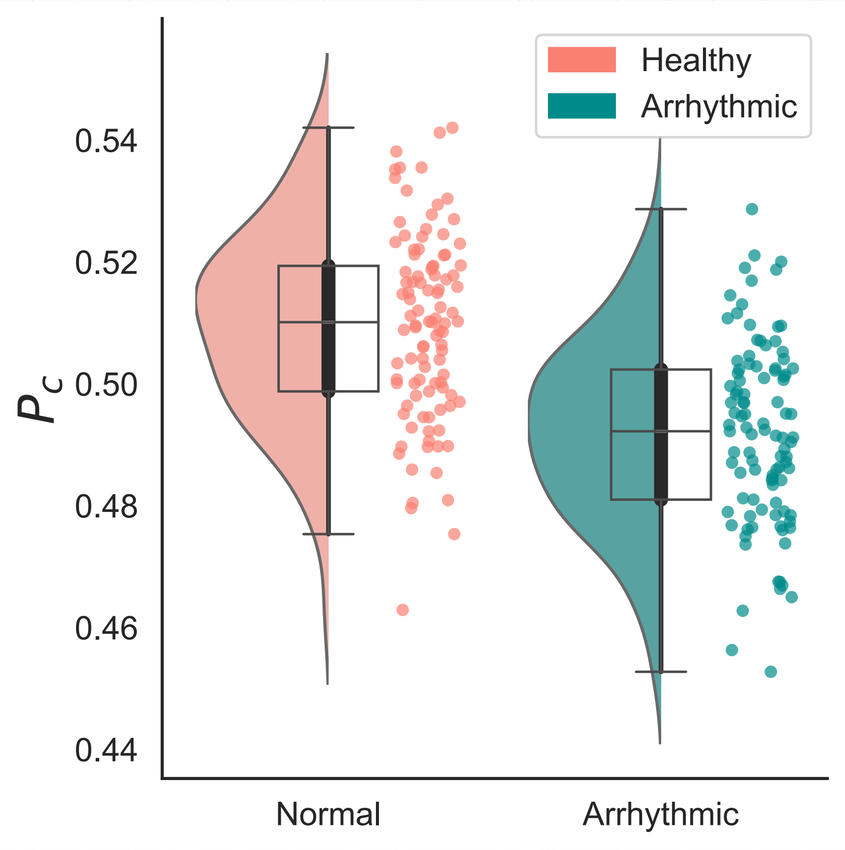




Healthy
Arrhythmic

Real-world data analysis

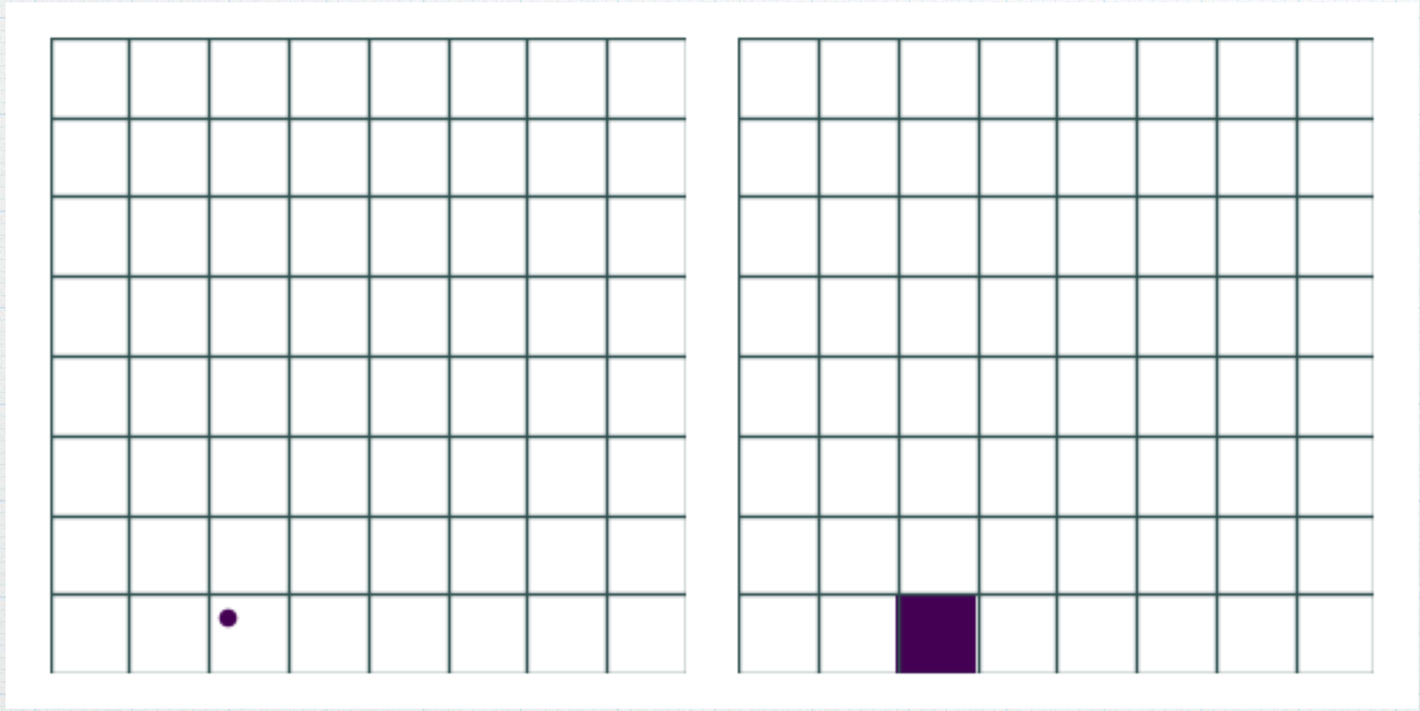


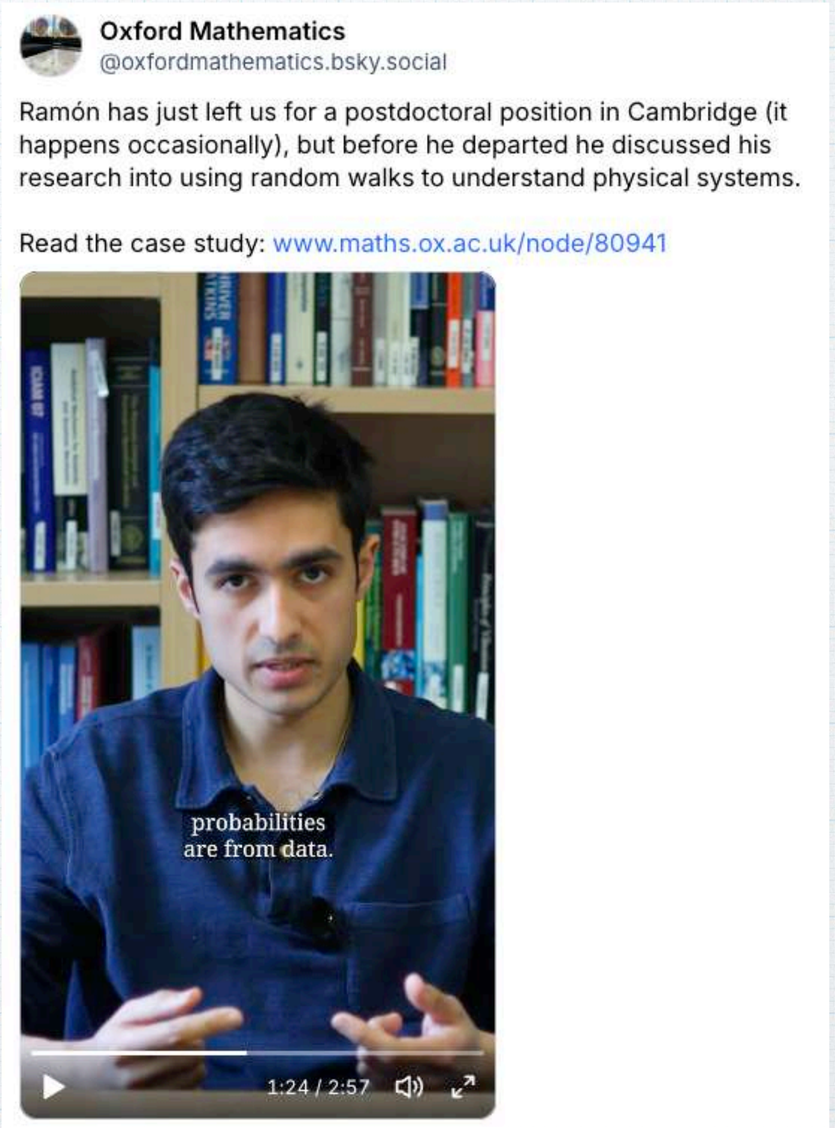


Using ensemble bootstrapping, we construct many ensembles of trajectories and find that irreversibility is significantly higher in healthy cells and healthy heartbeats, supporting previous literature results.

RNK et al. (2026) *JSTAT*

Follow up work:





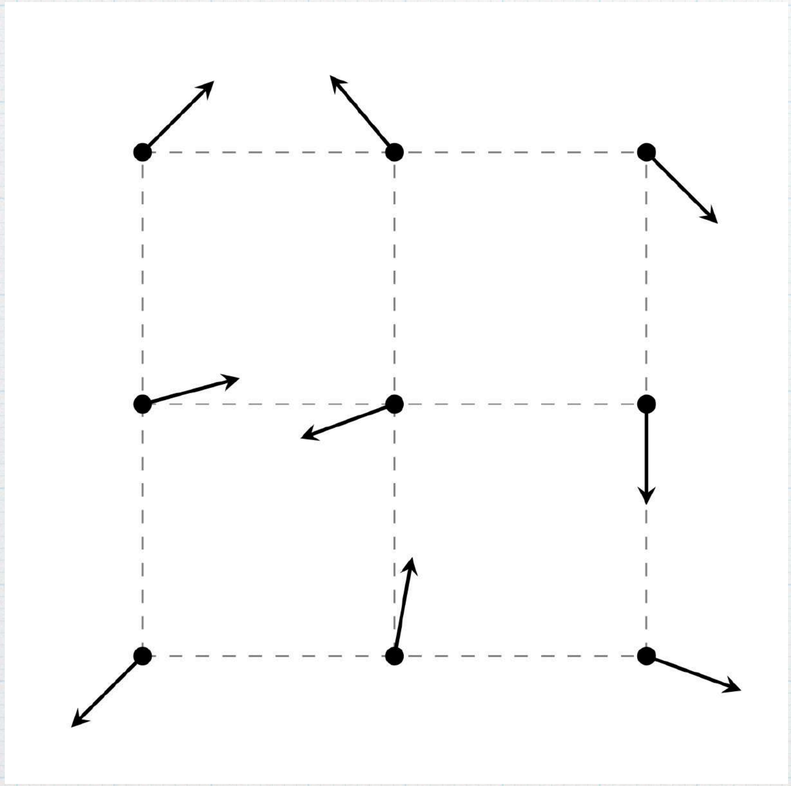
Project 2:
Detecting vortices in oscillator networks and neural dynamics

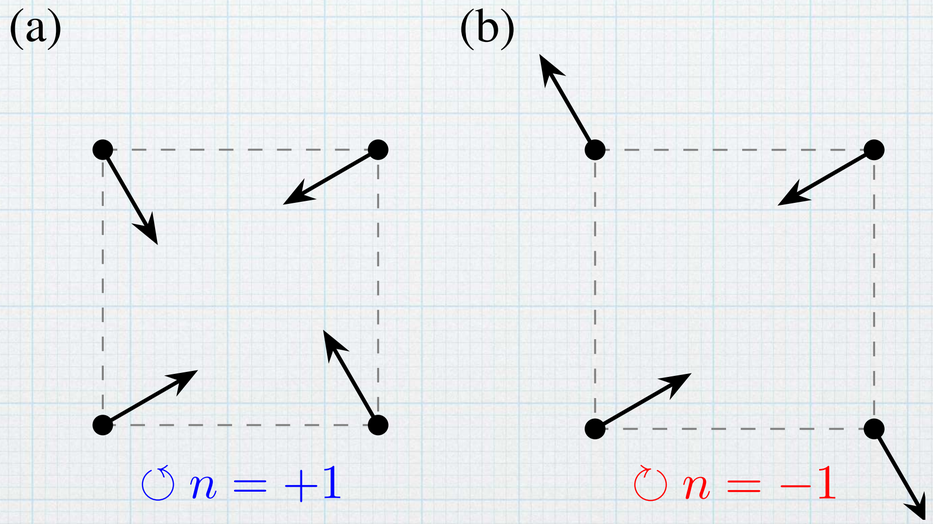
RNK*, Beers*, Guzman, Deco, Kringelbach, Lambiotte, Goriely, Porter (2026) Vortex-like patterns in oscillator networks and neuronal networks*; in prep*

Vortices

Phase oscillators on lattices can produce ‘vortices’: spiral-like patterns

$$\mathbf{s}_i = (\cos\theta_i, \sin\theta_i) \quad \theta_i \in [0,2\pi)$$





Chaikin & Lubensky (2012)

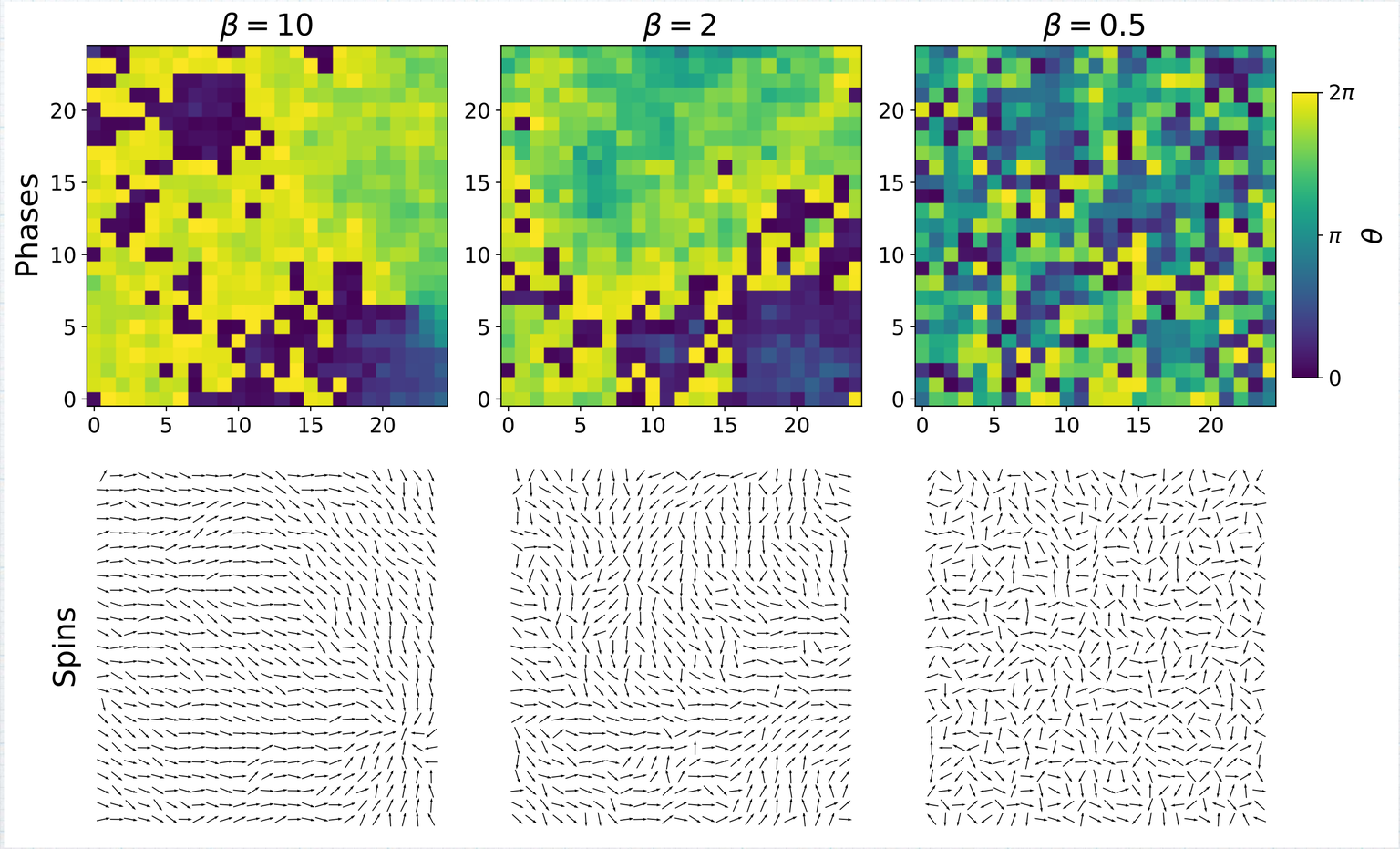
Vortices

The classic model that displays vortices is called the ‘XY’ model

$$\mathbf{s}_i = (\cos\theta_i, \sin\theta_i) \quad \theta_i \in [0,2\pi)$$

$$H(\{\mathbf{s}\}) = -\sum_{i,j} J_{ij}\cos(\theta_i - \theta_j)$$

$$P(\{\mathbf{s}\}) = \frac{e^{-\beta H(\{\mathbf{s}\})}}{Z}$$



Vortices

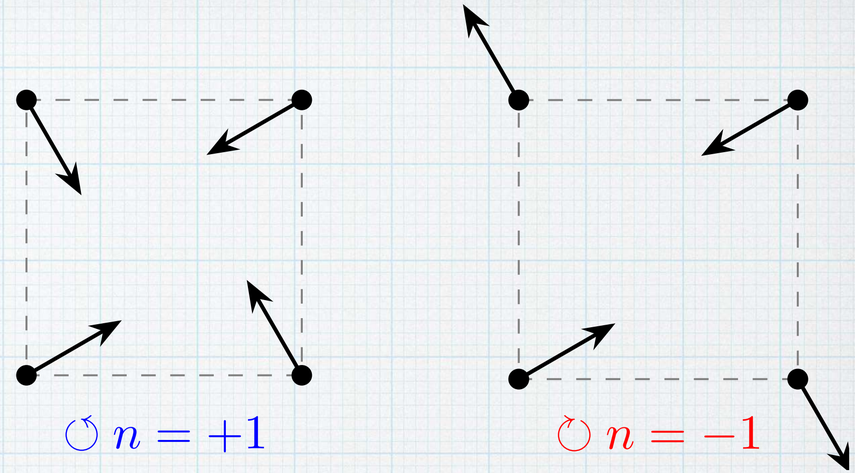
The classic model that displays vortices is called the ‘XY’ model

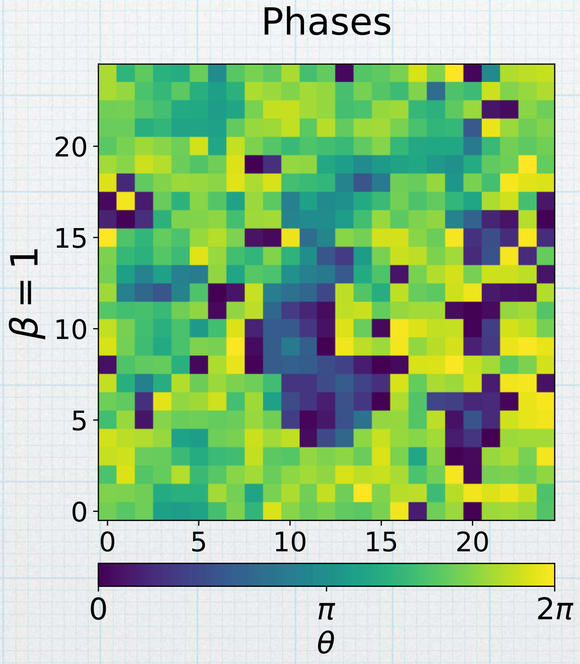
$$\mathbf{s}_i = (\cos\theta_i, \sin\theta_i) \quad \theta_i \in [0,2\pi)$$

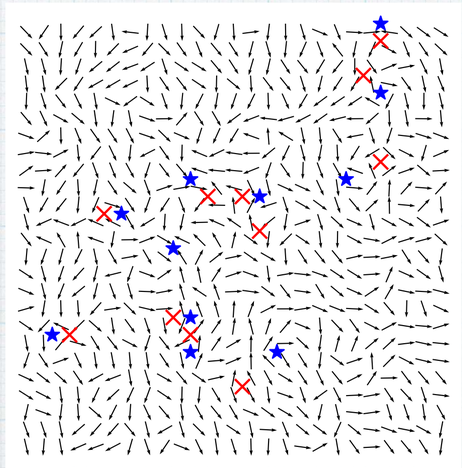
$$H(\{\mathbf{s}\}) = -\sum_{i,j} J_{ij}\cos(\theta_i - \theta_j)$$

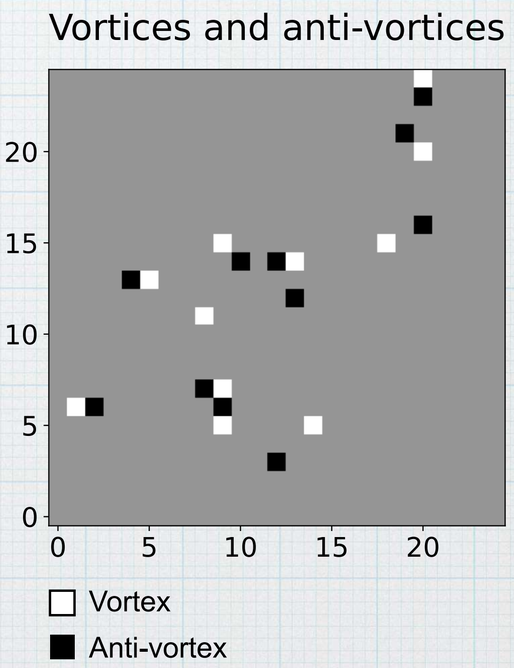
$$P(\{\mathbf{s}\}) = \frac{e^{-\beta H(\{\mathbf{s}\})}}{Z}$$

$$V = \sum_{i,j} \frac{|n_{i,j}|}{N}$$



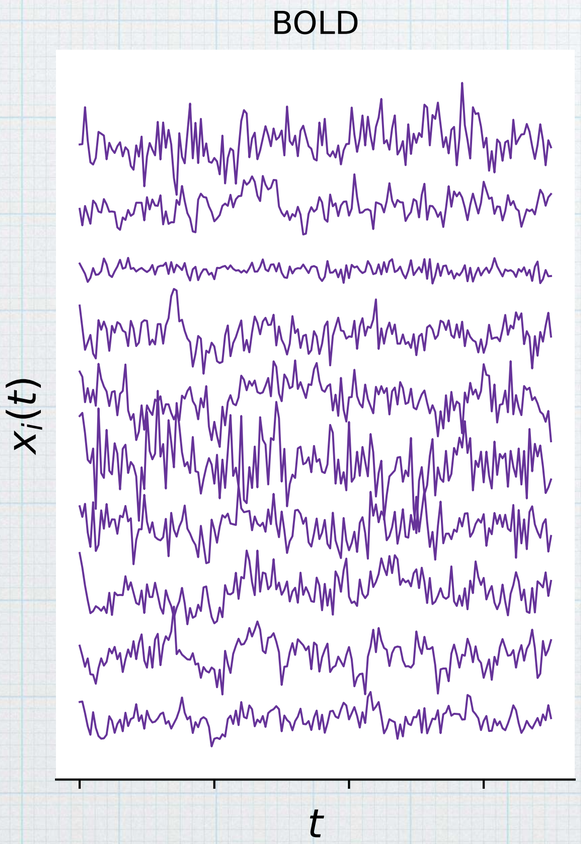


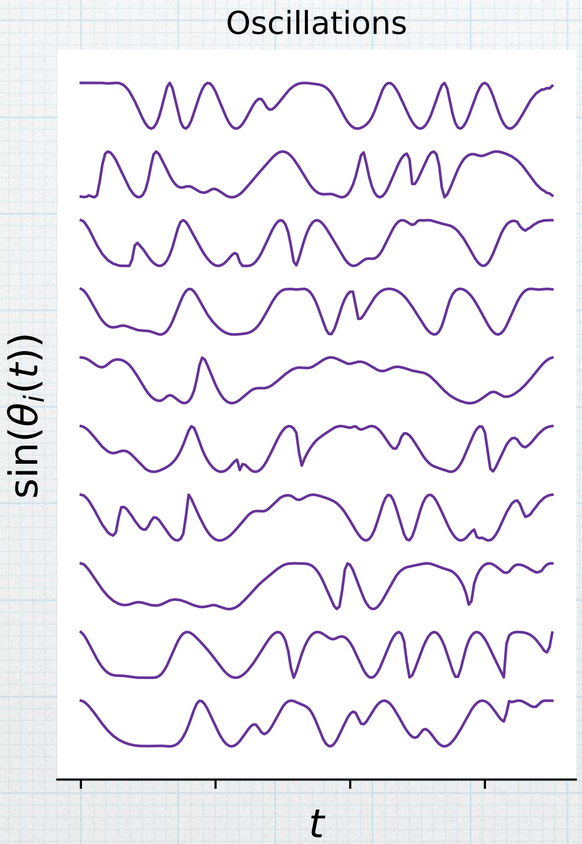


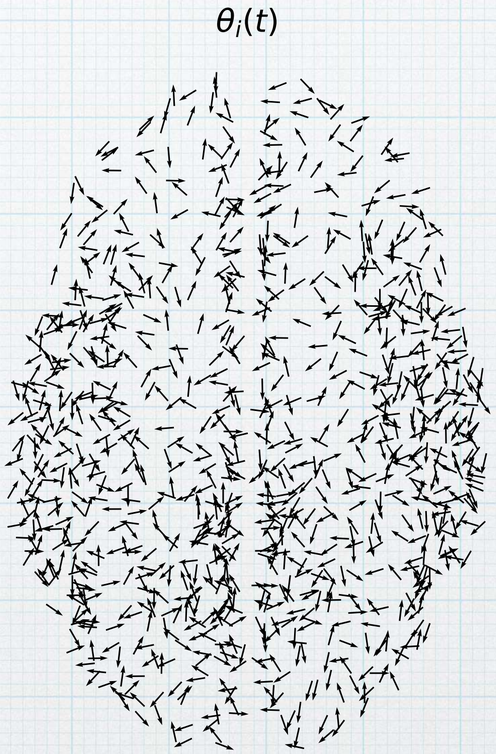


Vortices in the brain

There has been some recent interest in looking at vorticity in the brain, by looking at a flattened view of the cortex

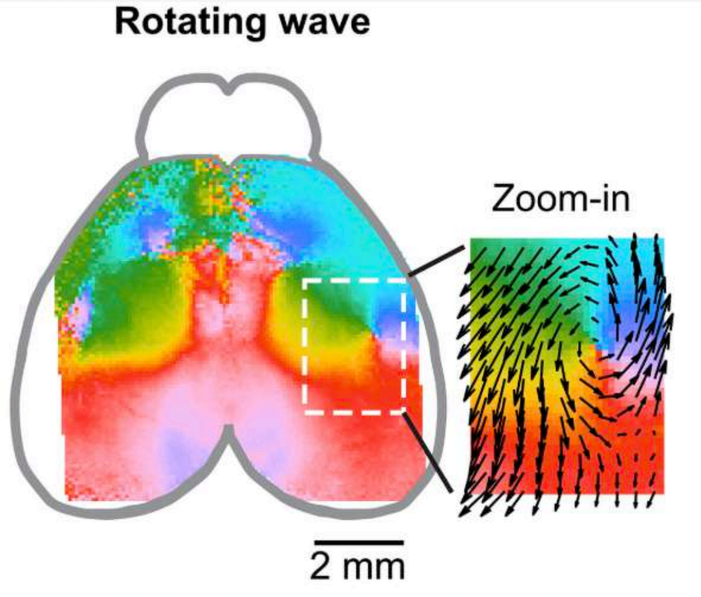




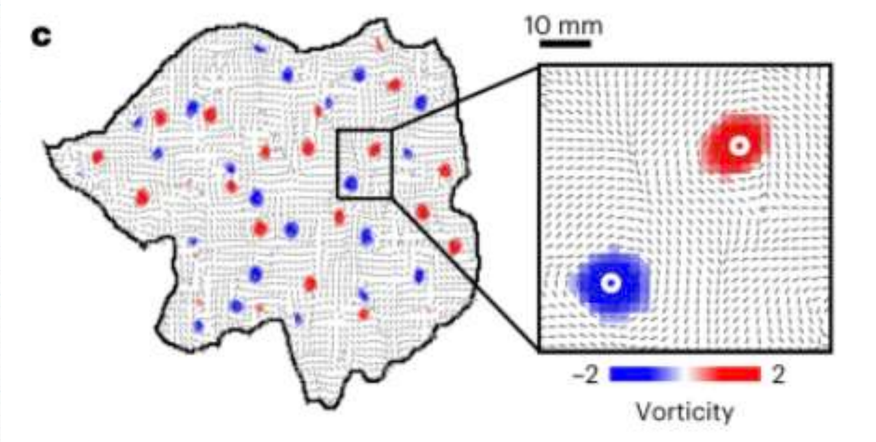


Vortices in the brain

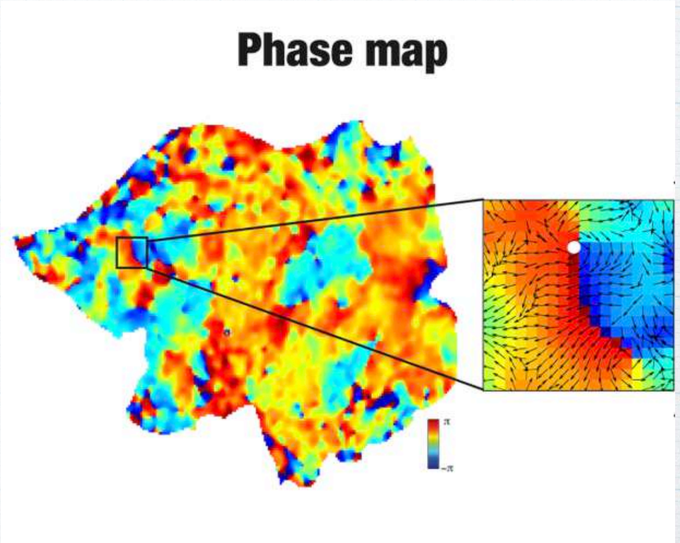
There has been some recent interest in looking at vorticity in the brain, by looking at a flattened view of the cortex



Ye et al. (2026) *Science*



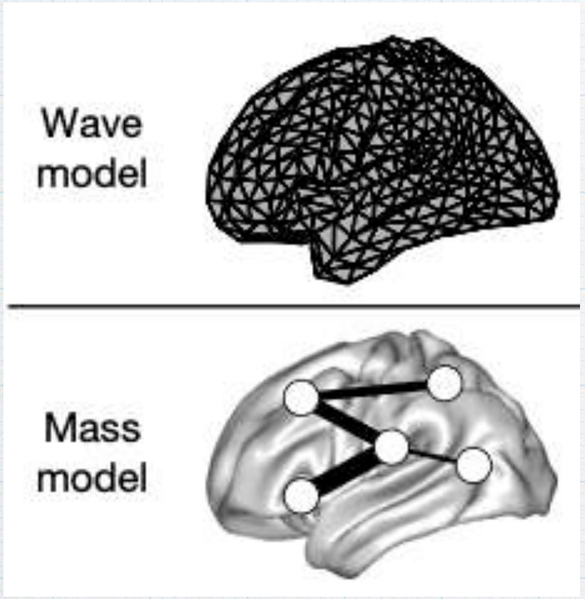
Xu et al. (2023) *Nat. Hum. Beh.*



Deco et al. (2026) *Net. Neuro.*

Topological vs geometric brain

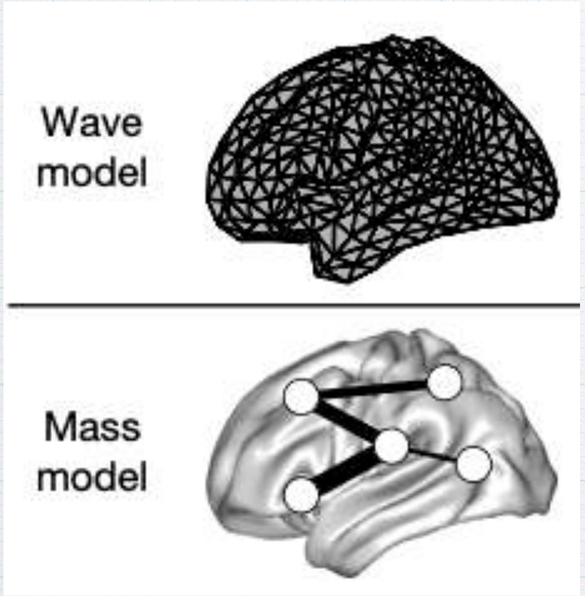
Is the brain's geometry or topology more important for *structure-function coupling*?



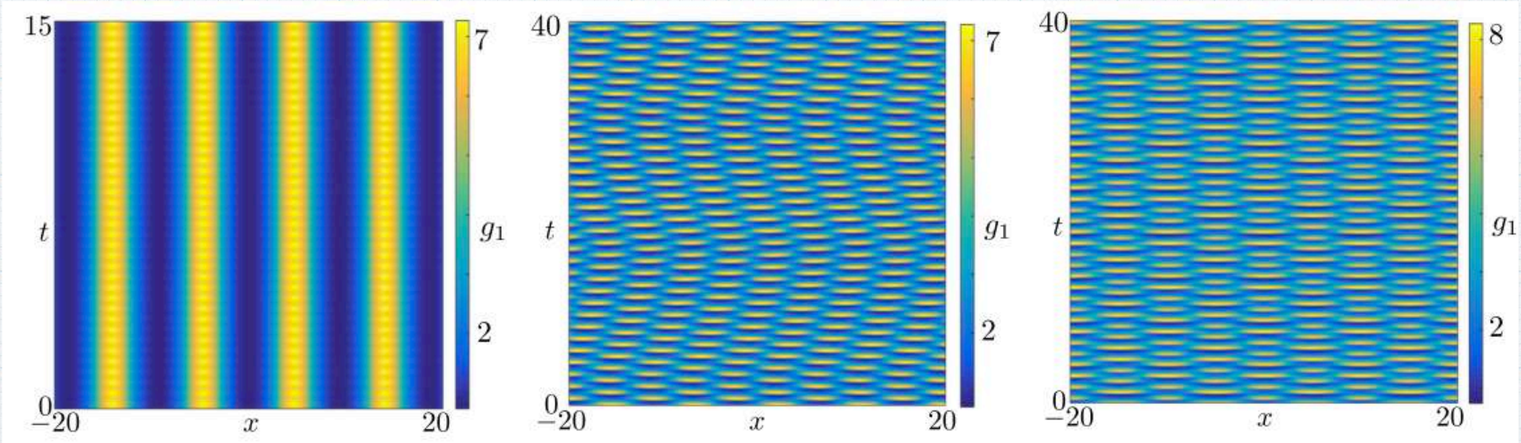
Pang et al. (2023) *Nature*

Topological vs geometric brain

Is the brain's geometry or topology more important for *structure-function coupling*?



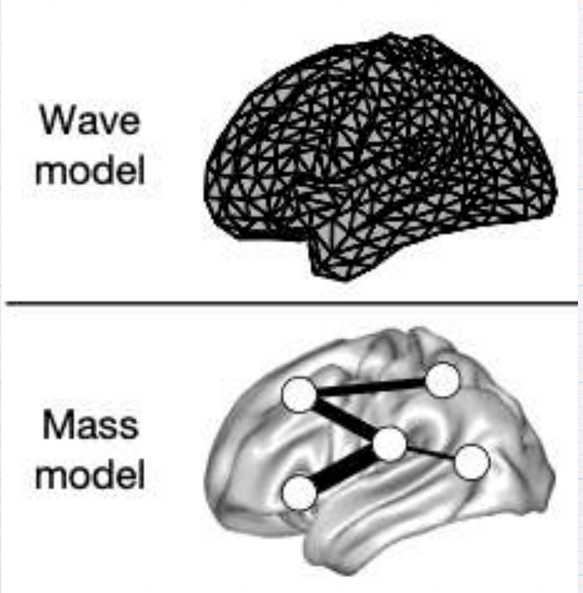
Pang et al. (2023) *Nature*



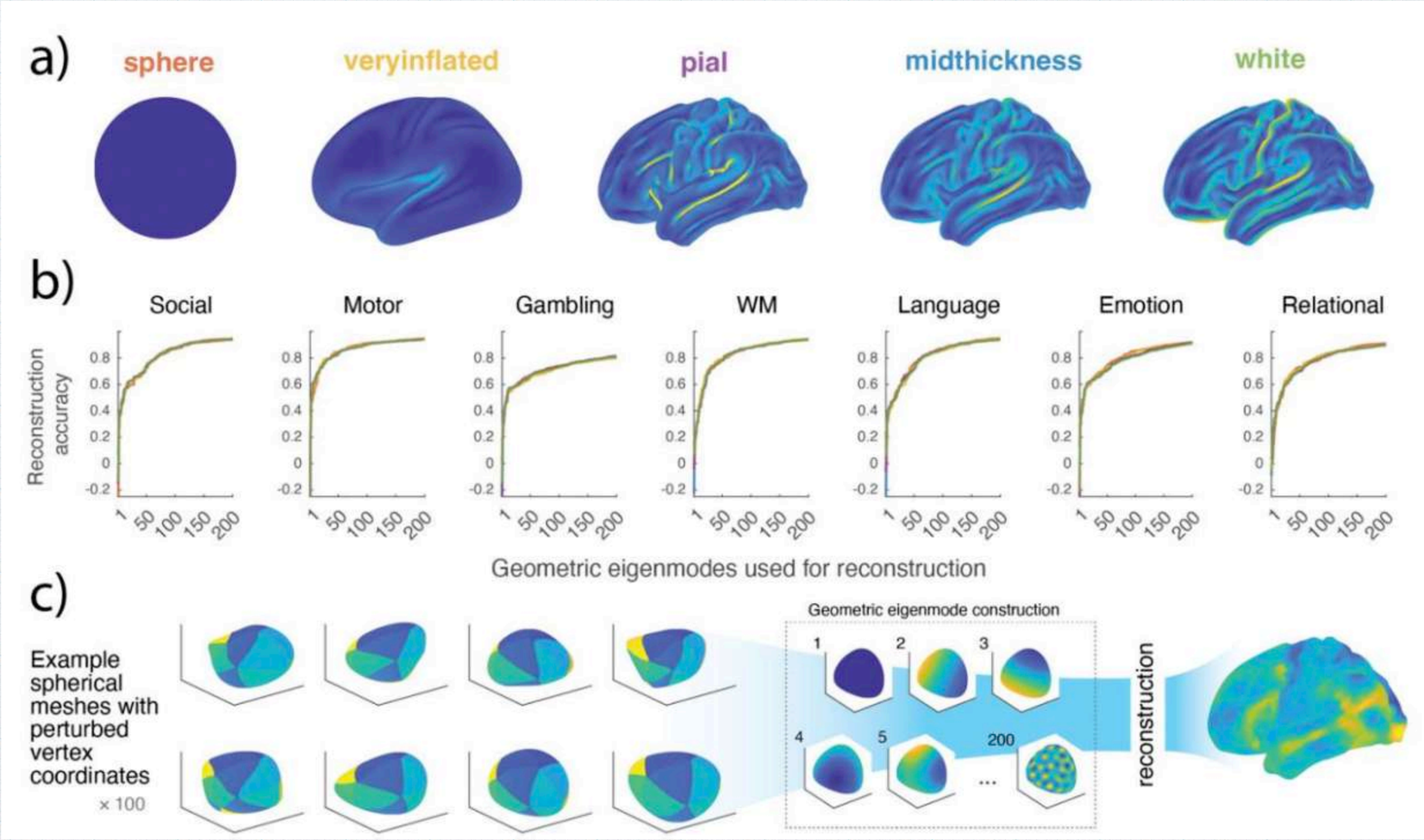
Byrne et al. (2019) *Phy. Rev. E*

Topological vs geometric brain

Is the brain's geometry or topology more important for *structure-function coupling*?



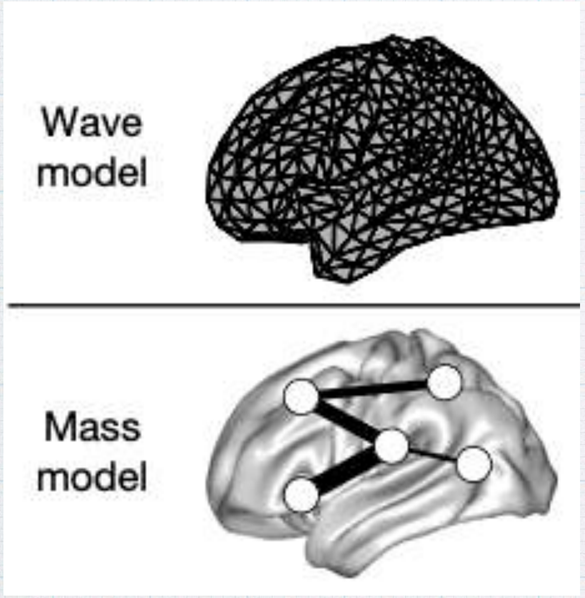
Pang et al. (2023) *Nature*



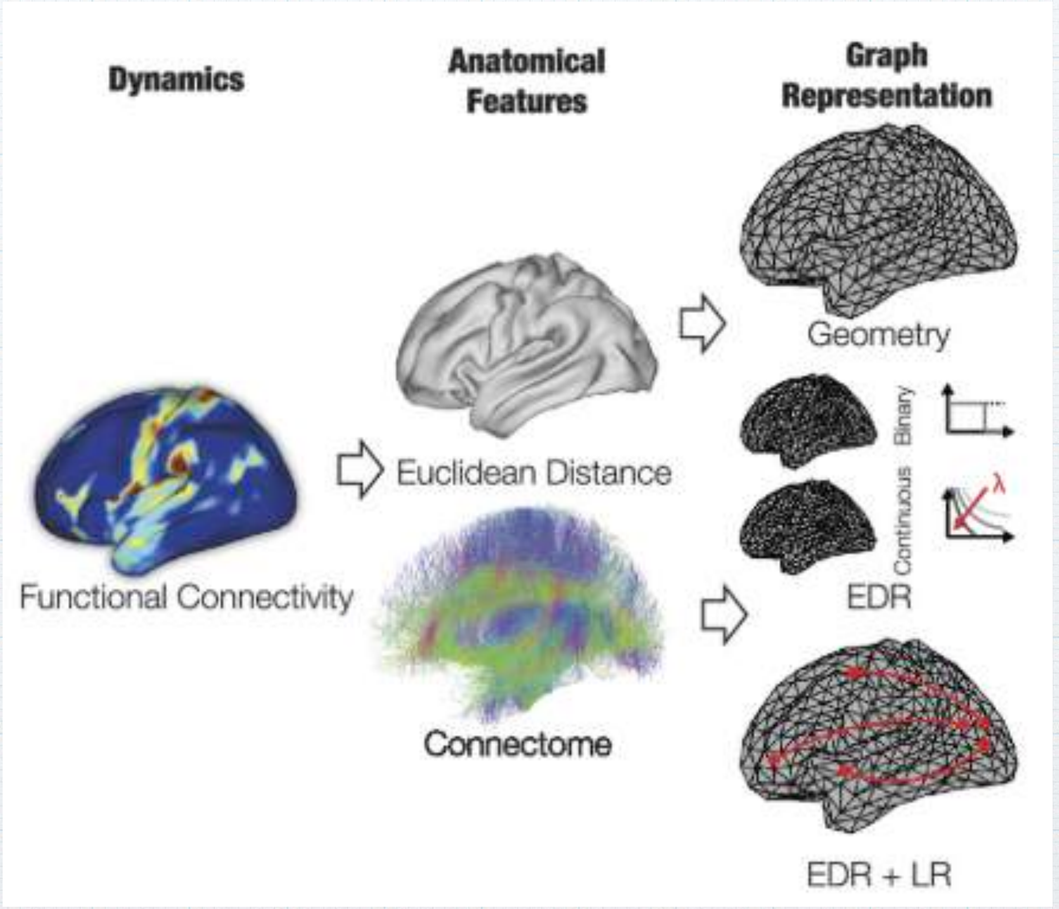
Faskowitz et al. (2023) *bioRxiv*

Topological vs geometric brain

Is the brain's geometry or topology more important for *structure-function coupling*?

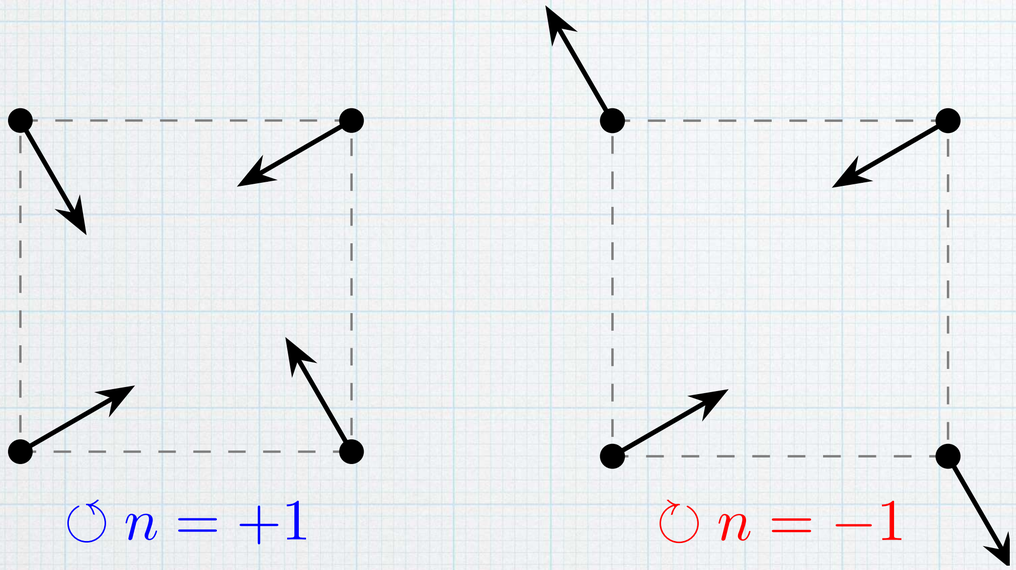


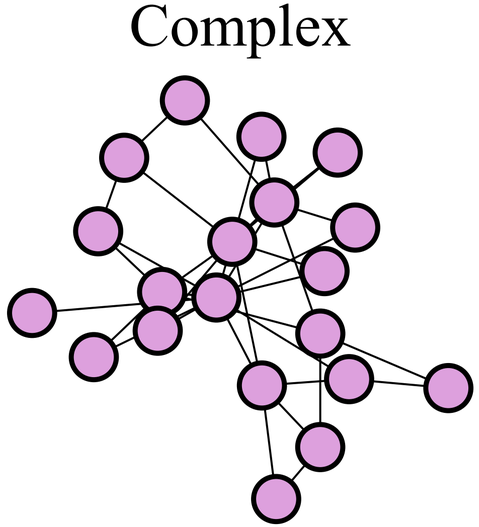
Pang et al. (2023) *Nature*

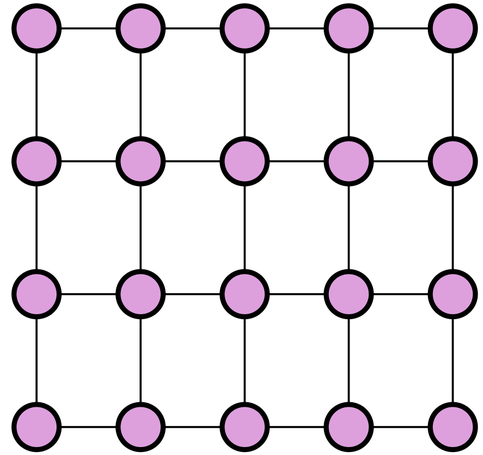


Vohryzek et al. (2025) *PNAS*

How do you identify vortices on a network?







Use the Hodge decomposition!

$$X(i,j) = \text{wrap}(\theta_j - \theta_i)$$

$$\text{wrap}(\alpha) = \alpha - 2\pi \cdot \text{round}\left(\frac{\alpha}{2\pi}\right)$$

$$X(i,j) = (\text{grad}\, f)(i,j) + R(i,j)$$



David Beers, UCLA

Use the Hodge decomposition!

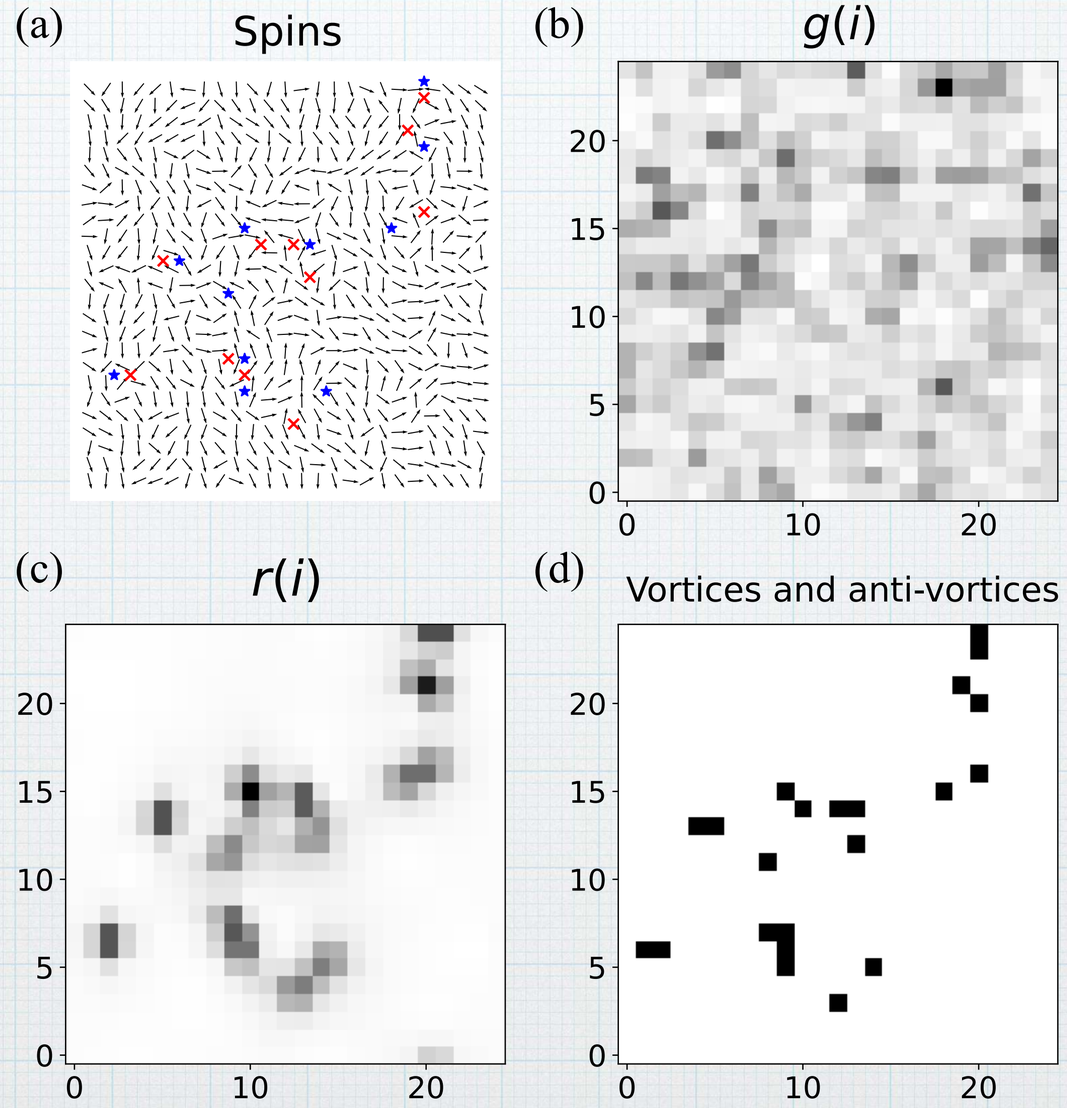
$$X(i,j) = \text{wrap}(\theta_j - \theta_i)$$

$$\text{wrap}(\alpha) = \alpha - 2\pi \cdot \text{round}\left(\frac{\alpha}{2\pi}\right)$$

$$X(i,j) = (\text{grad}\, f)(i,j) + R(i,j)$$

$$r(i) = \sum_{j:(i,j)\in E} |R(i,j)|,$$

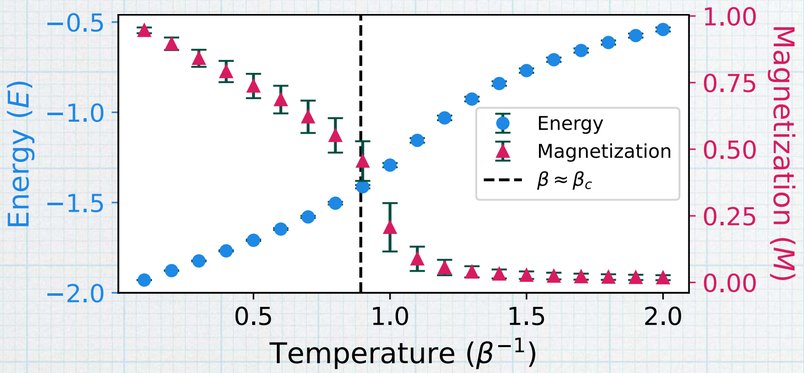
$$g(i) = \sum_{j:(i,j)\in E} |(\text{grad}\, f)(i,j)|.$$

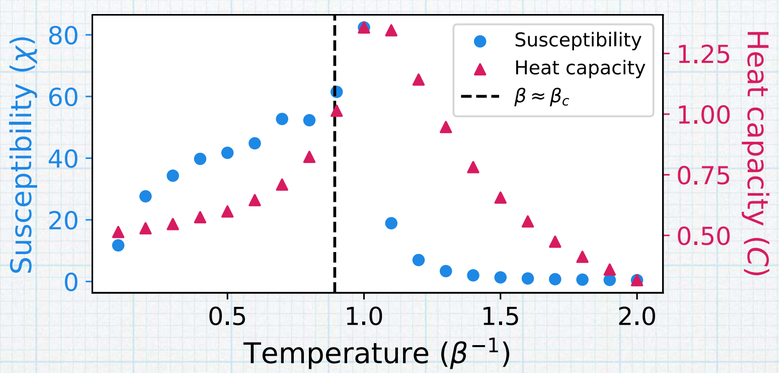


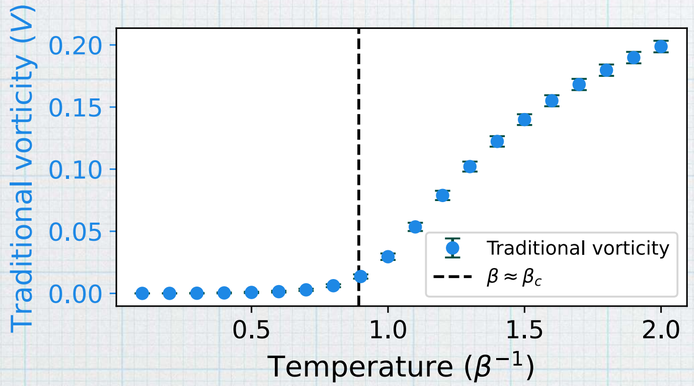


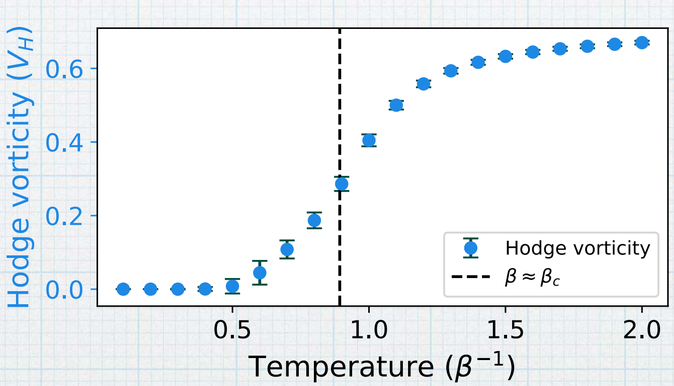
David Beers, UCLA

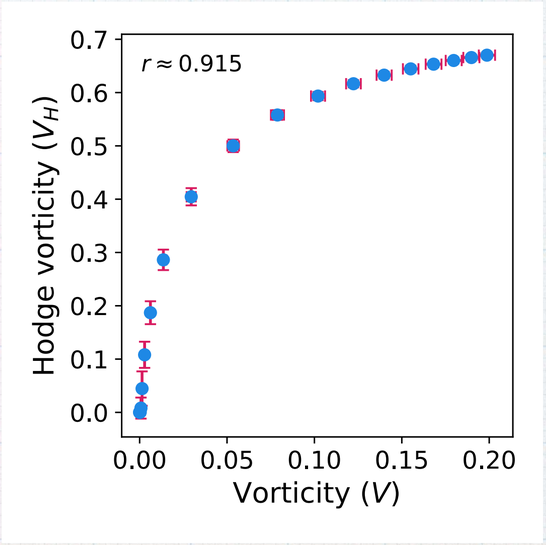
Numerical results: the critical point on the lattice







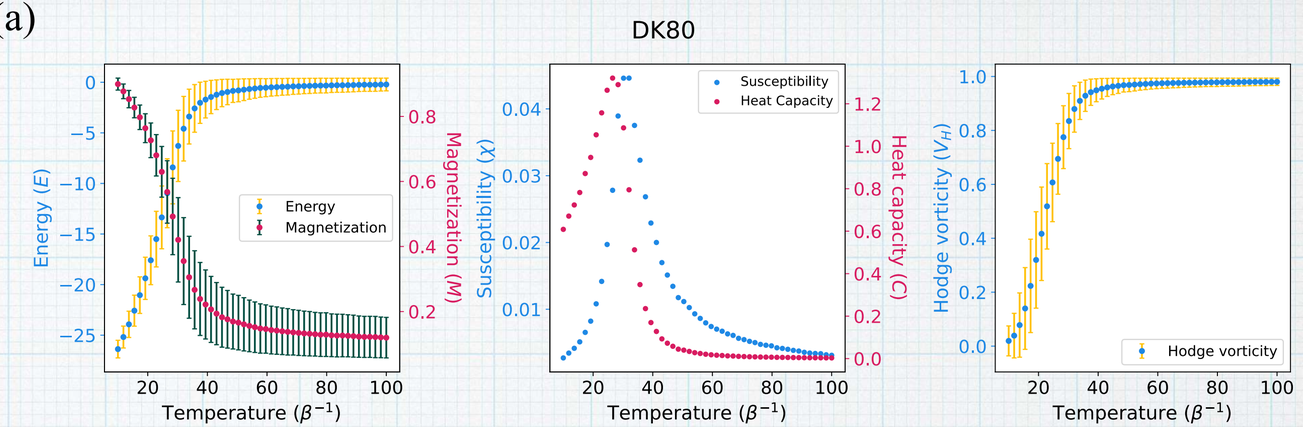




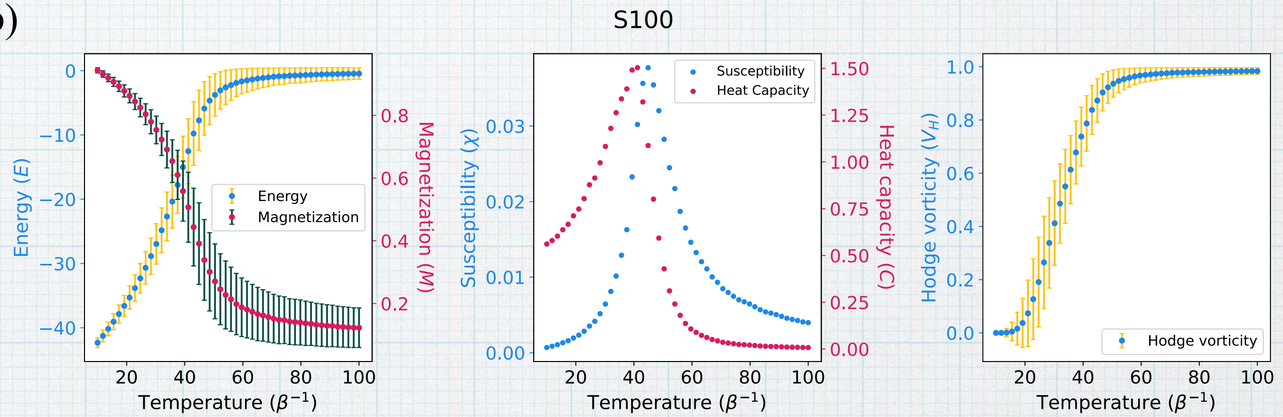
Kosterlitz & Thouless (1972) *Journal of Physics C*

Numerical results: the critical point on connectomes

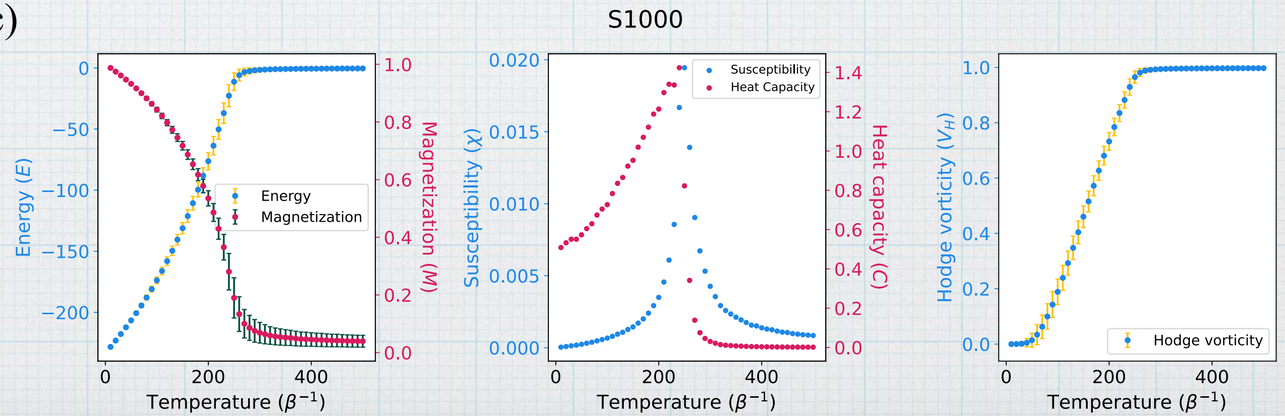
(a)



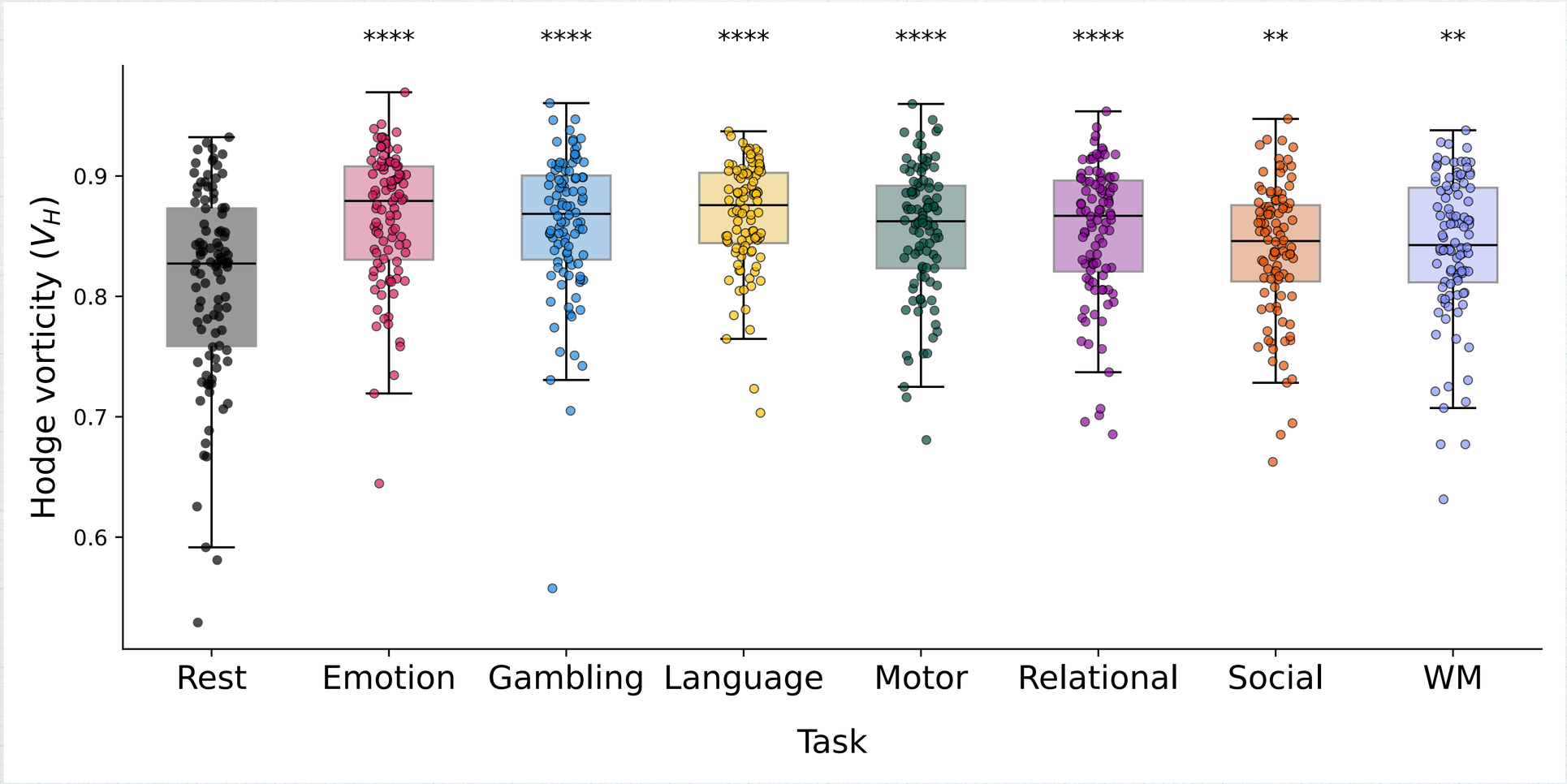
(b)



(c)

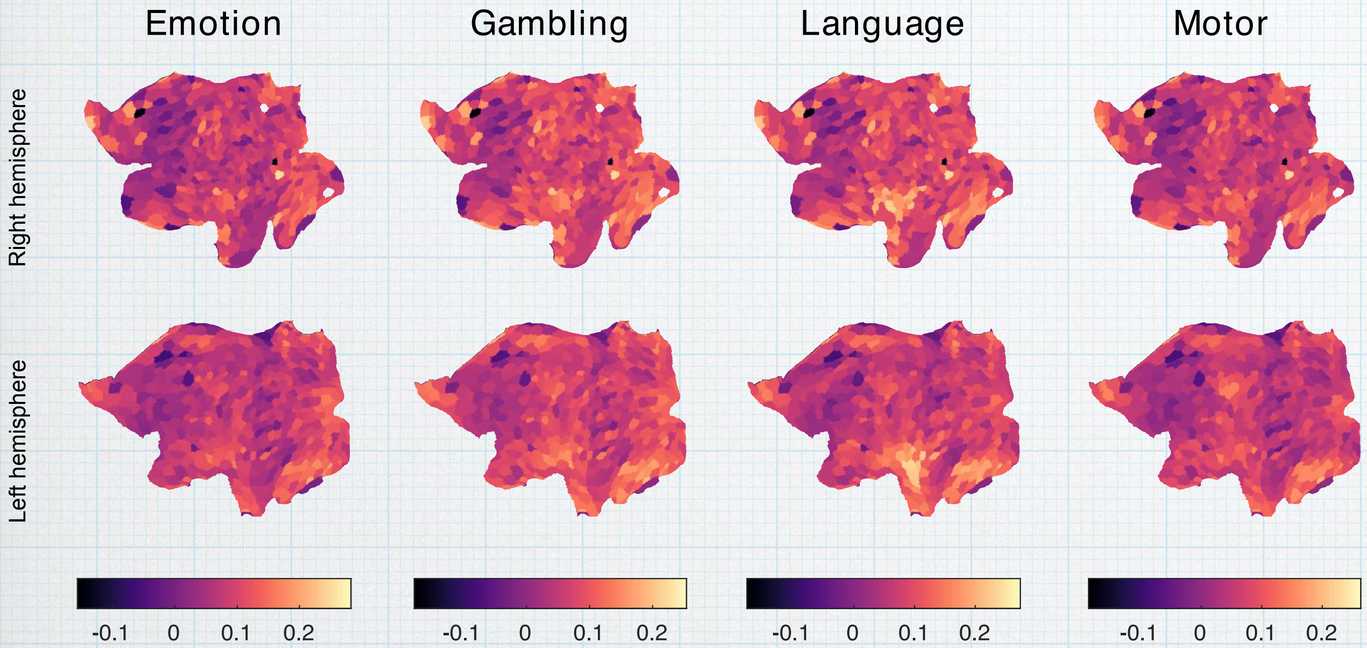


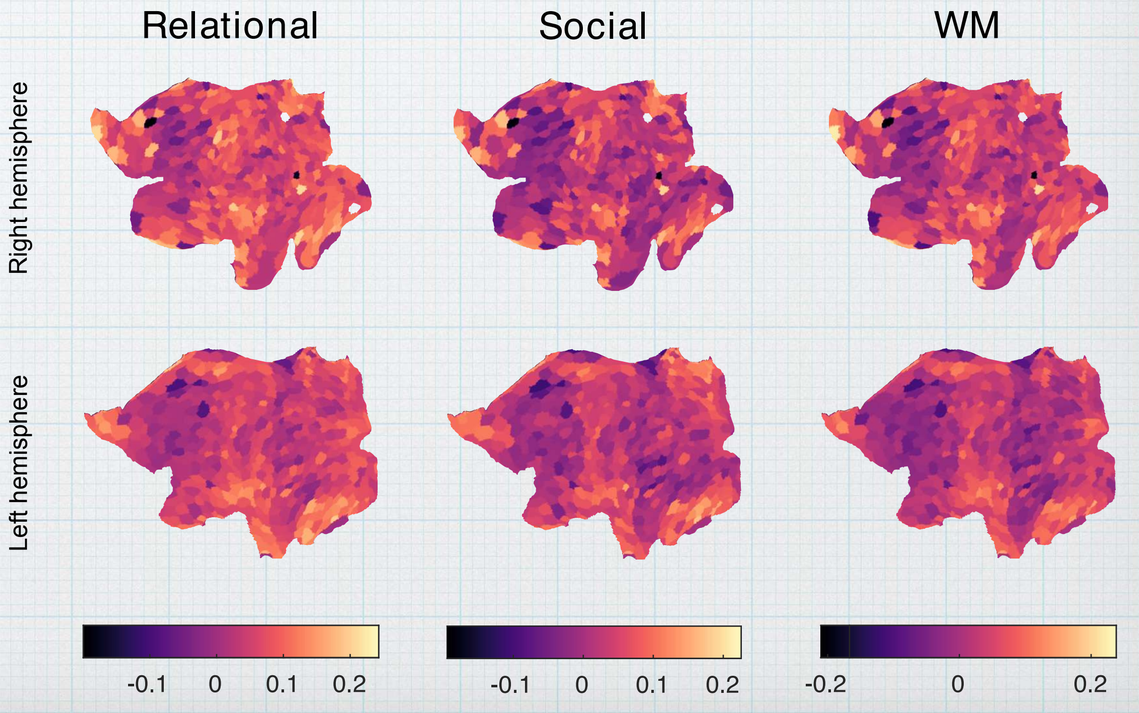
Vorticity in tasks and resting -state fMRI



100 unrelated participants; Schaefer 1000; [0.008,0.08] Hz

Vorticity in tasks and resting -state fMRI

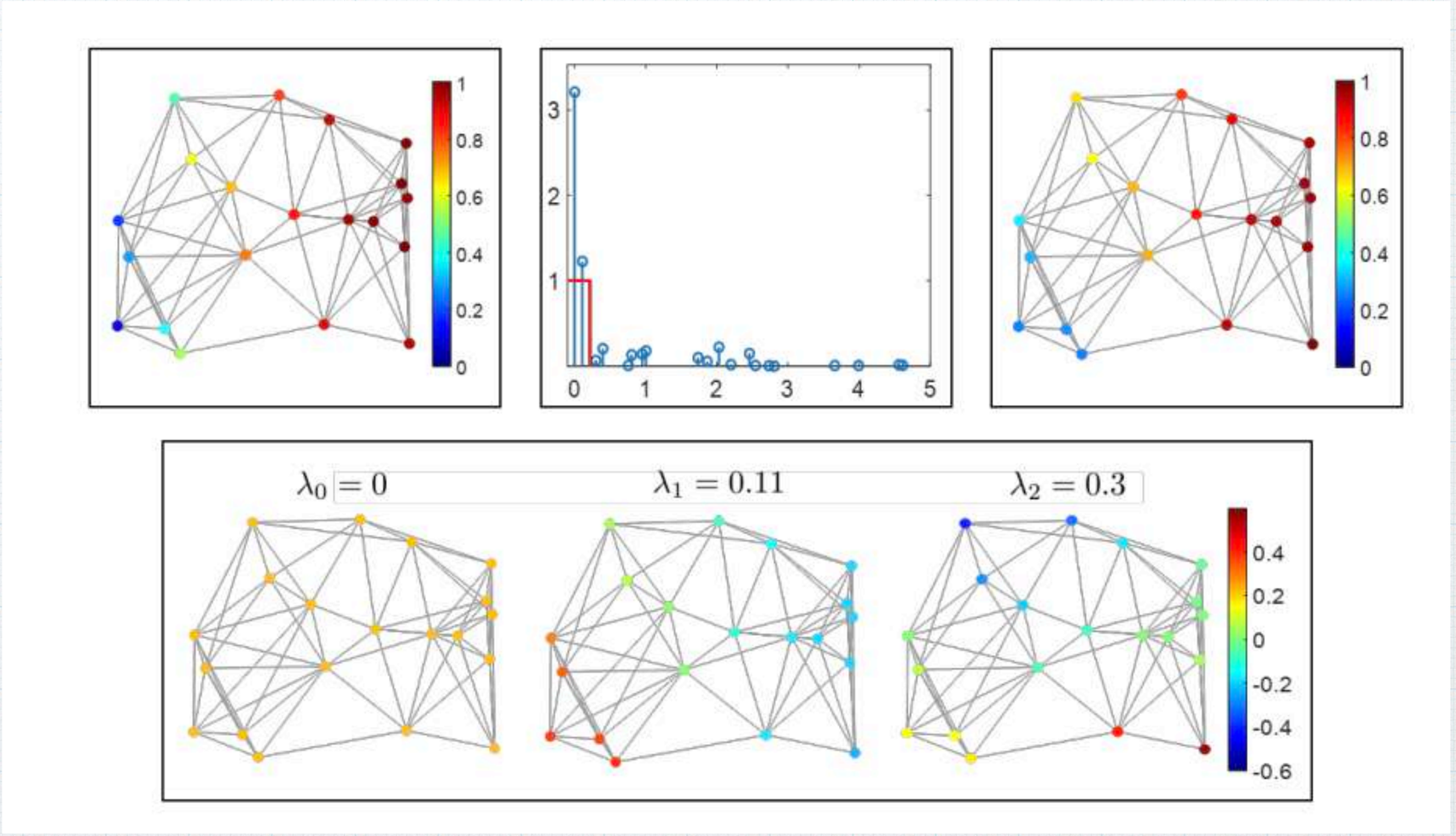




Eigenmodes of the Laplacian

Signal processing with the graph Laplacian is an area of significant interest

$$L\psi = \lambda\psi$$



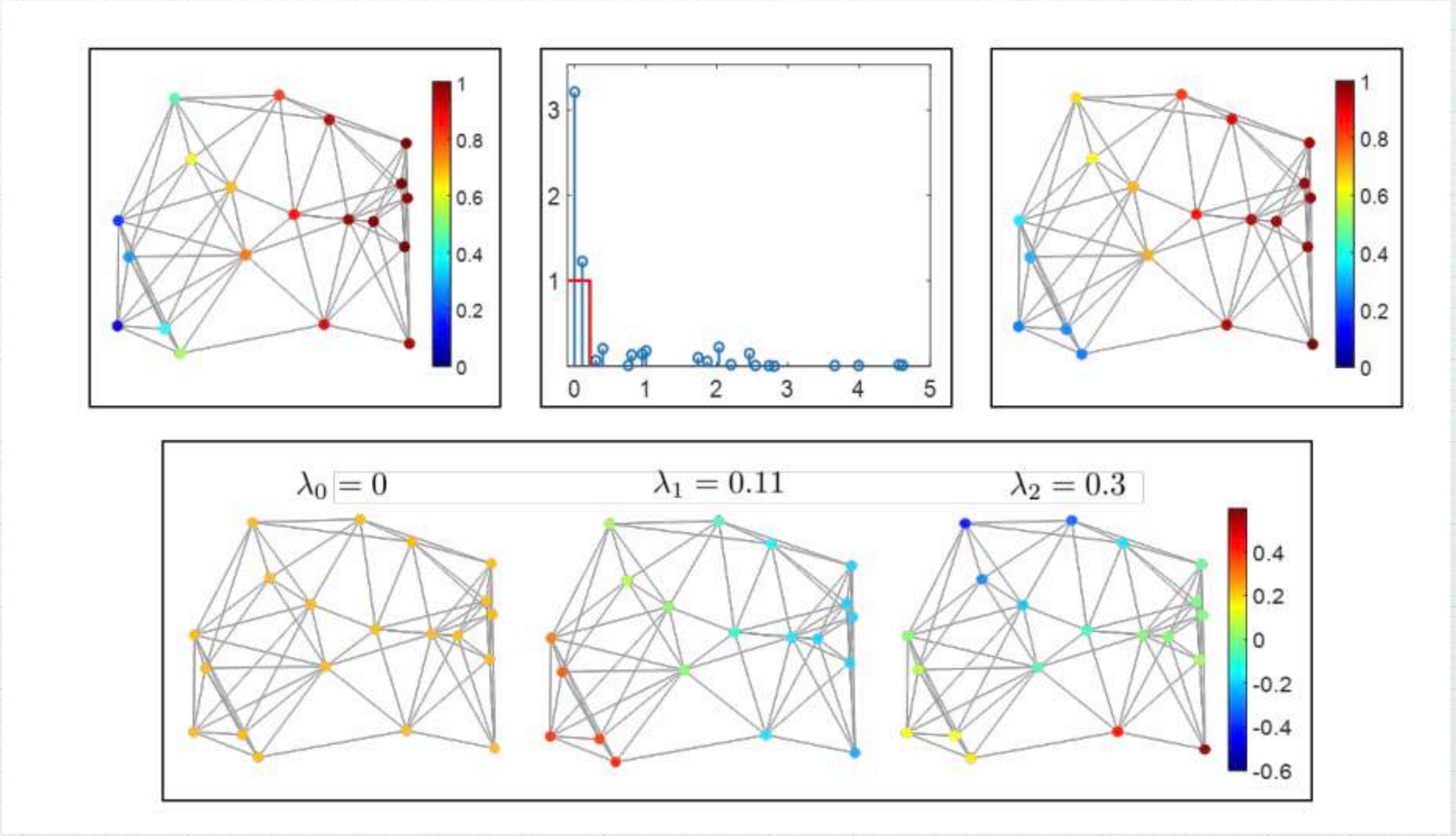
Leus et al. (2023) *IEEE Signal Processing Magazine*

Eigenmodes of the Laplacian

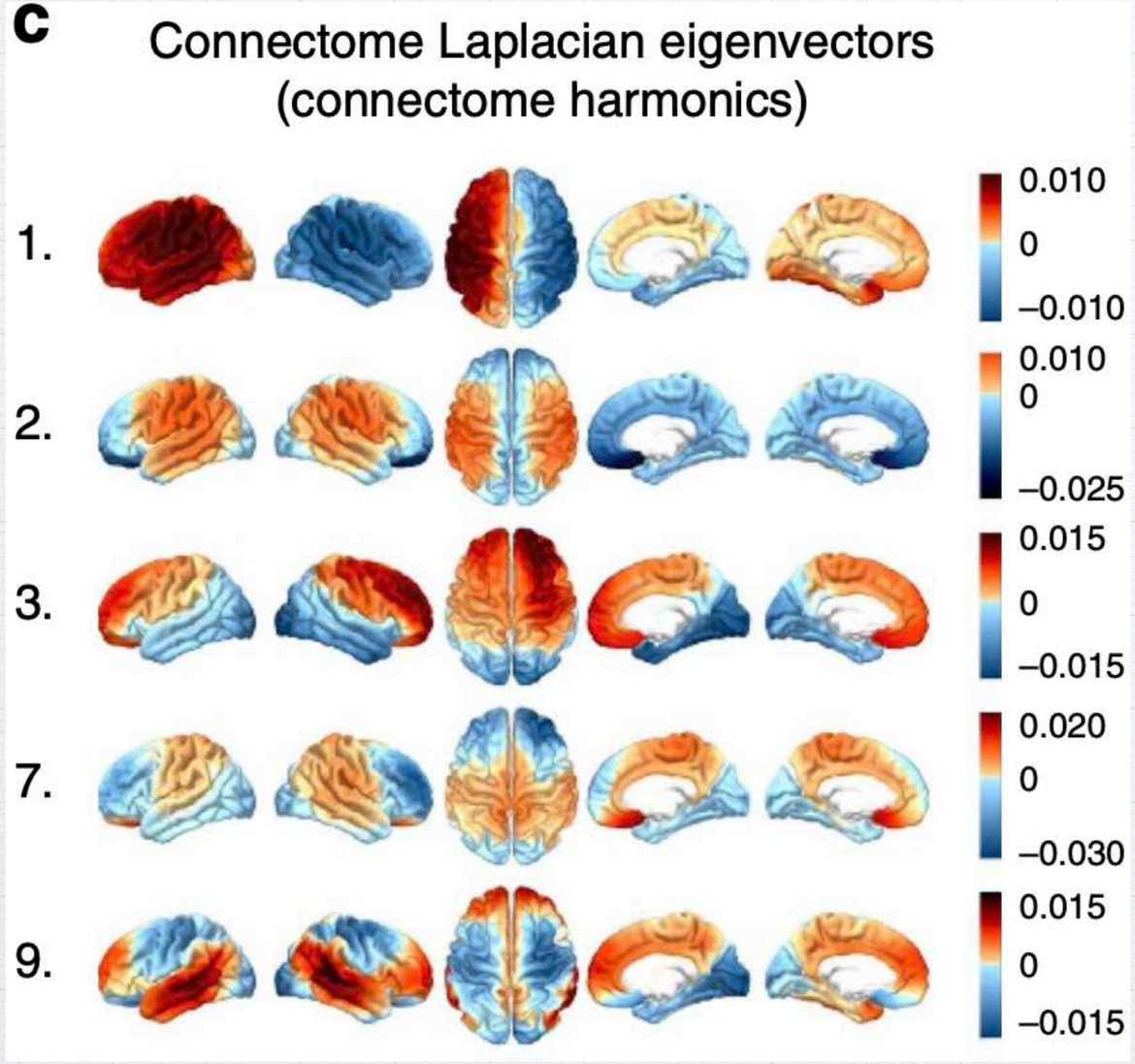
Signal processing with the graph Laplacian is an area of significant interest

$$L\psi = \lambda\psi$$

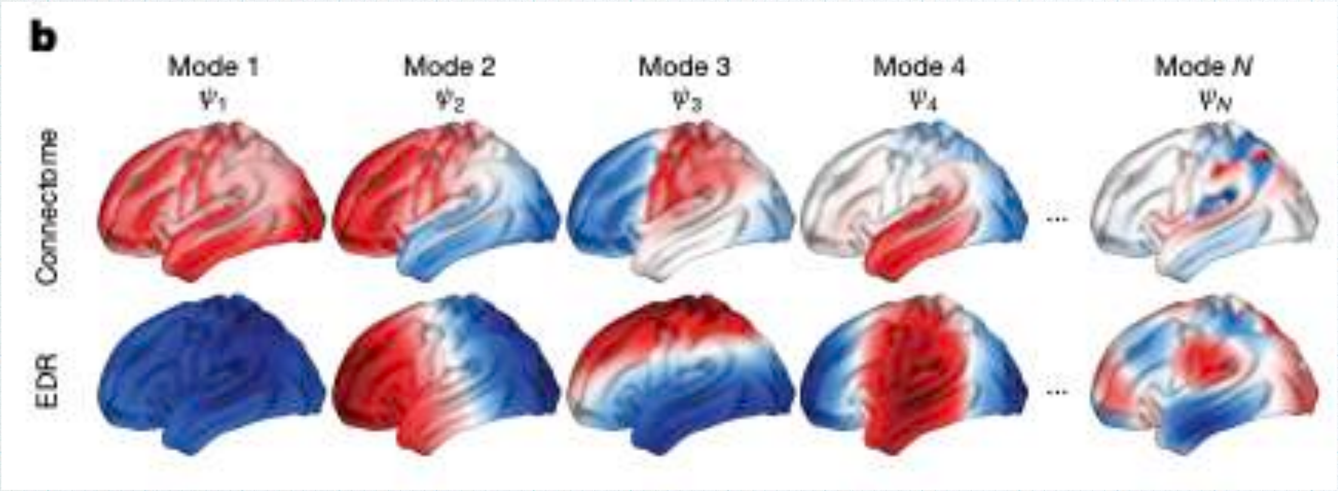
$$x(t) = \sum_{j=1}^{M} c_j(t)\psi_j$$

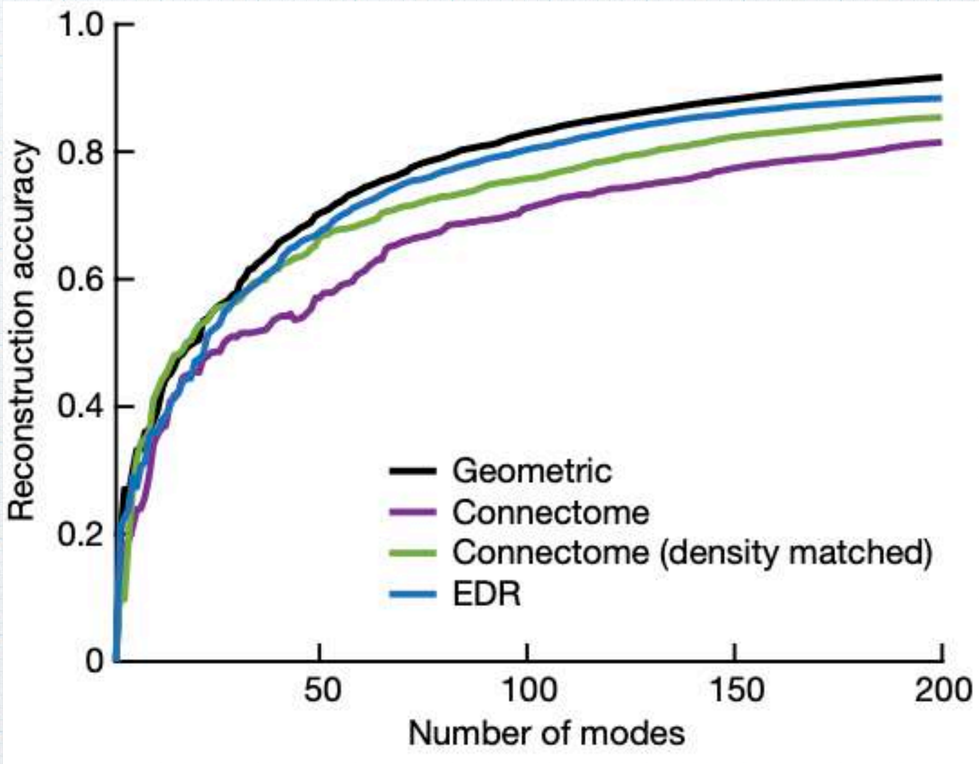


Leus et al. (2023) *IEEE Signal Processing Magazine*



Atasoy et al. (2016) *Nat. Comms.*

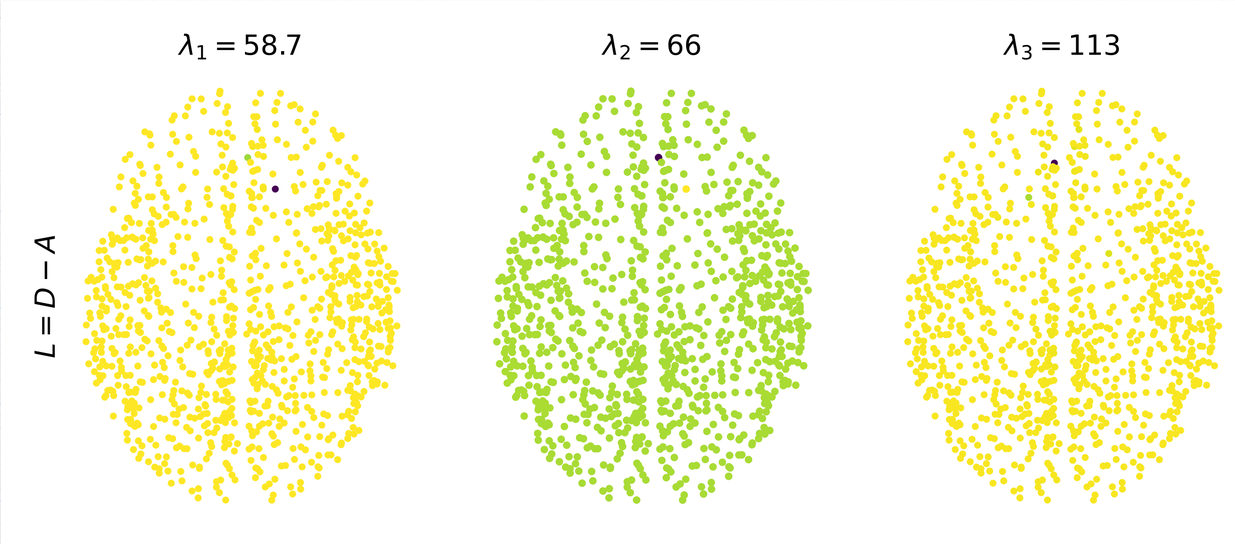




Pang et al. (2023) *Nature*

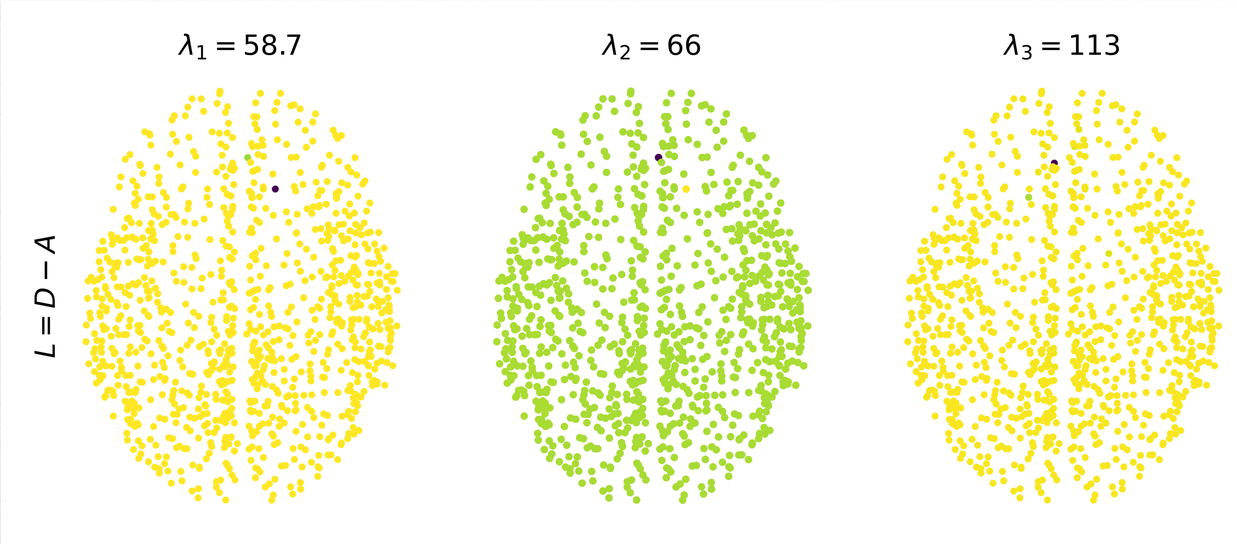
Eigenmodes of the Laplacian

Caveat: you have to use the **normalised** Laplacian or the modes are skewed by **high-degree nodes**



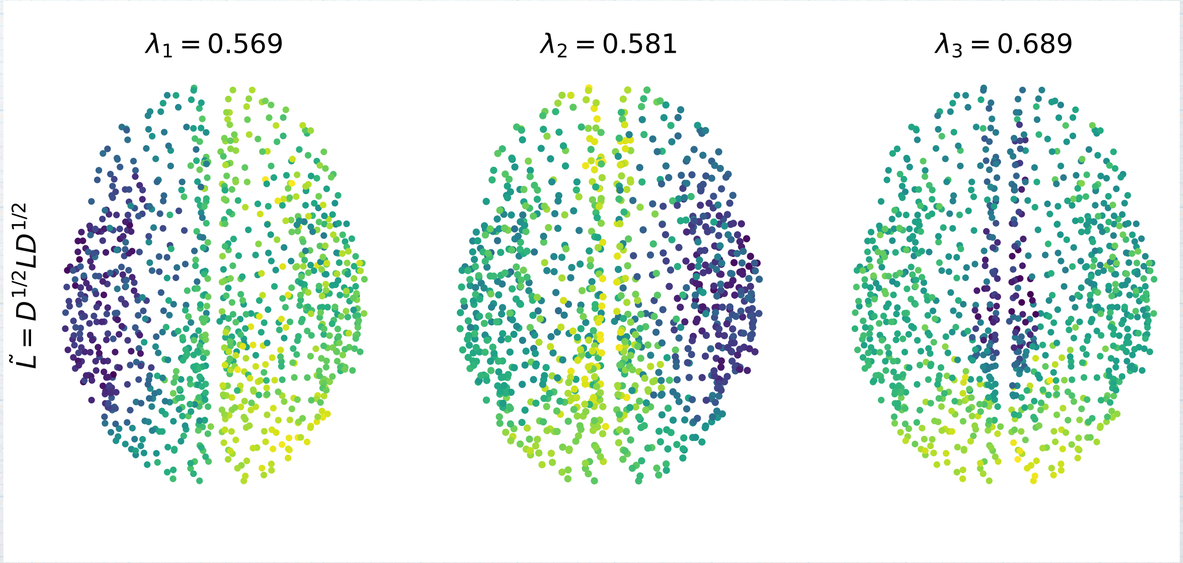
Eigenmodes of the Laplacian

Caveat: you have to use the **normalised** Laplacian or the modes are skewed by **high-degree nodes**



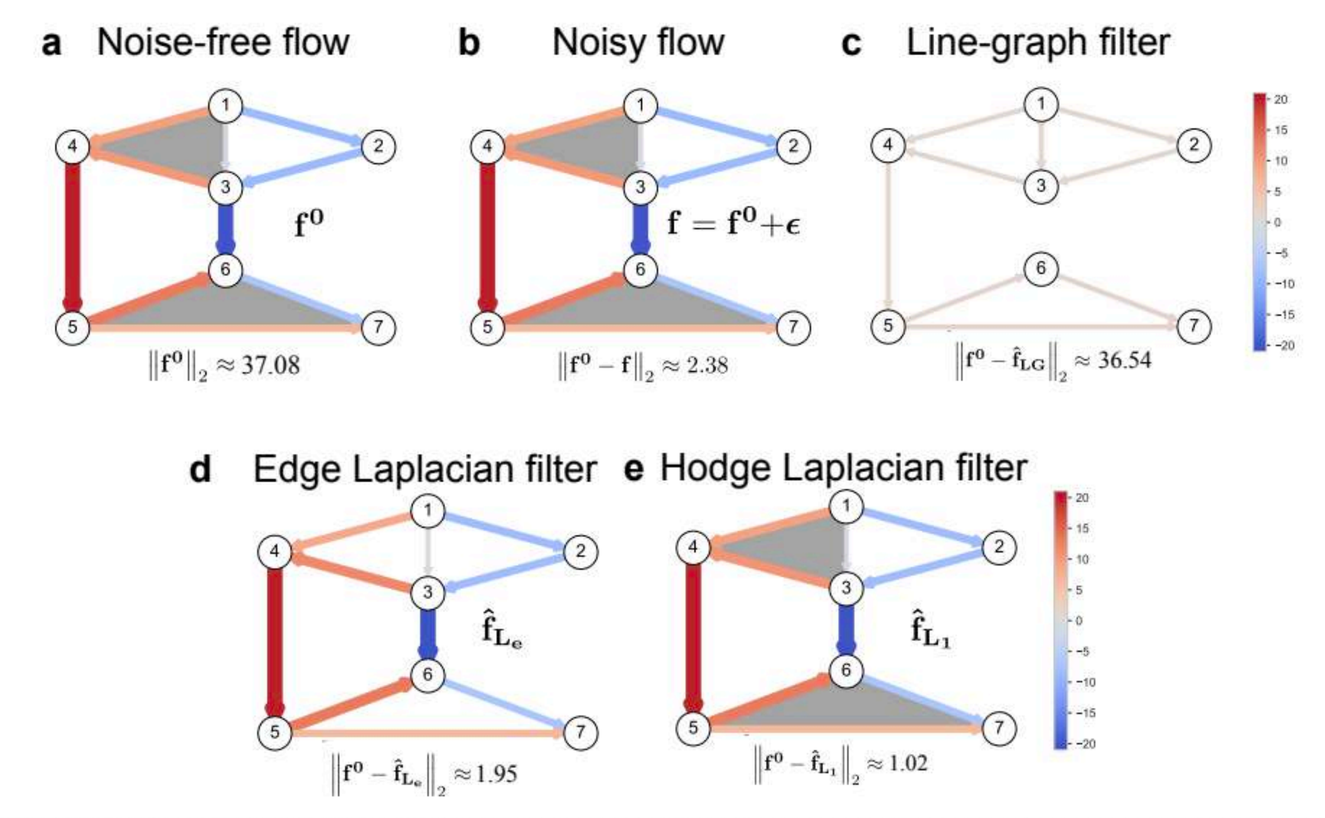
This is called **localisation**

Martin et al. (2014) *Phy. Rev. E*



Eigenmodes of the Hodge Laplacian

We can do signal processing of **edge-based** signals using the Hodge Laplacian



Schaub et al. (2022) *in Higher Order Systems*

Eigenmodes of the Hodge Laplacian

We can also do this for neural signals because we have defined an edge-flow

But do we need to do a normalisation?

Do we have this localisation in the Hodge Laplacian?

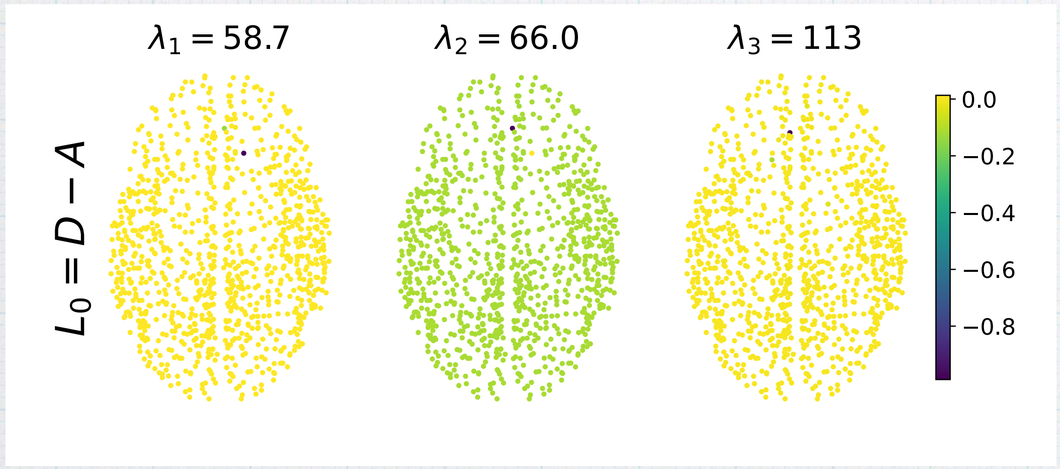
Eigenmodes of the Hodge Laplacian

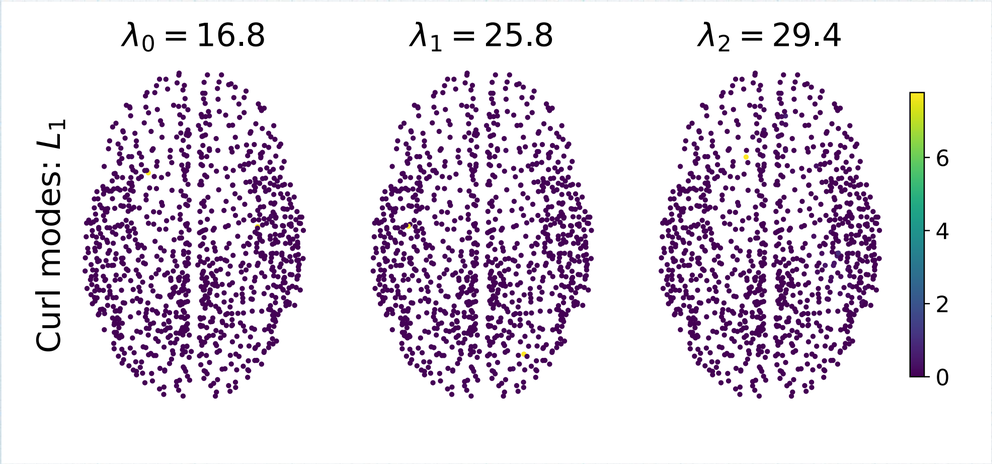
We can also do this for neural signals because we have defined an edge-flow

But do we need to do a normalisation?

Do we have this localisation in the Hodge Laplacian?

Yes!



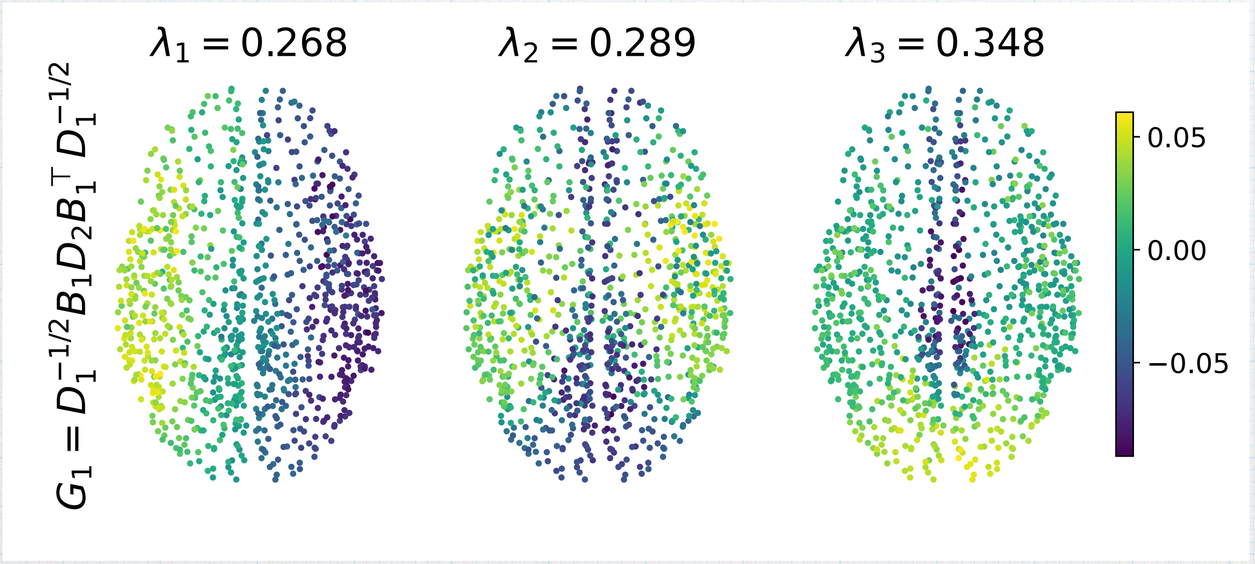


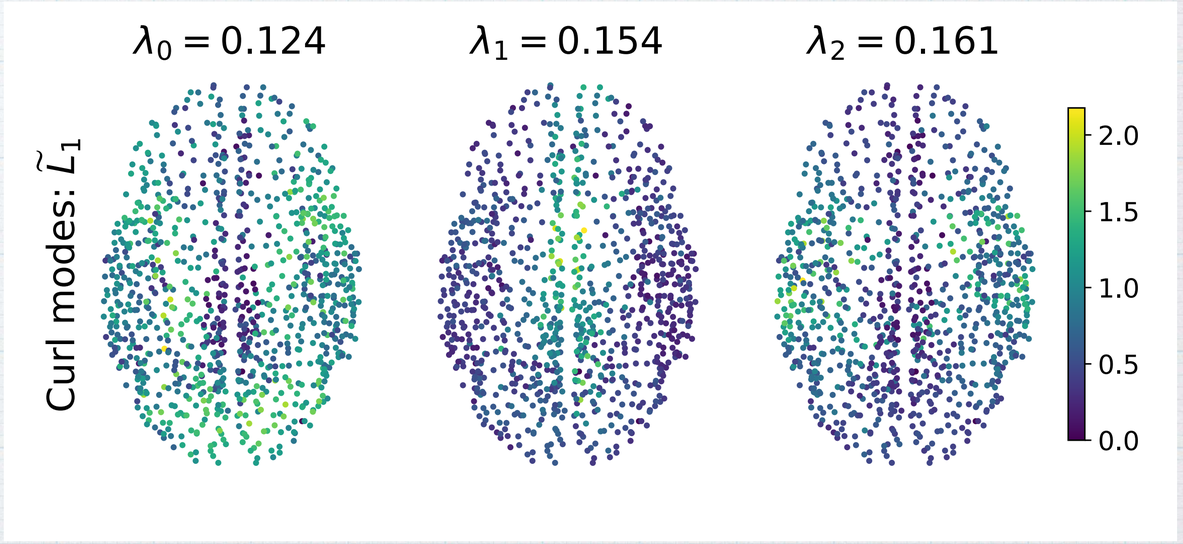
Eigenmodes of the Hodge Laplacian

We can also do this for neural signals because we have defined an edge-flow

But do we need to do a normalisation?

Do we have this localisation in the Hodge Laplacian?





Schaub et al. (2020) *SIAM Review*

Eigenmodes of the Hodge Laplacian

$$X(i,j) = \mathrm{wrap}(\theta_j - \theta_i)$$

$$X(t) = \sum_{j=1}^{M_1} a_j(t)\xi_j + \sum_{j=1}^{M_2} b_j(t)\psi_j \qquad L_1 = B_1^\top B_1 + B_2 B_2^\top$$

Eigenmodes of the Hodge Laplacian

$$X(i,j) = \text{wrap}(\theta_j - \theta_i)$$

$$X(t) = \sum_{j=1}^{M_1} a_j(t)\xi_j + \sum_{j=1}^{M_2} b_j(t)\psi_j \qquad L_1 = B_1^\top B_1 + B_2 B_2^\top$$

$$E_U(k) = \langle |\Upsilon_U^k(t)|^2 \rangle_t,$$

$$P_U(k) = \frac{E_U(k)}{\sum_{l=1}^{Q} E_U(l)}.$$

$$\mathcal{E}_U = -\sum_{k=1}^{Q} P_U(k) \log\left(P_U(k)\right),$$

Eigenmodes of the Hodge Laplacian

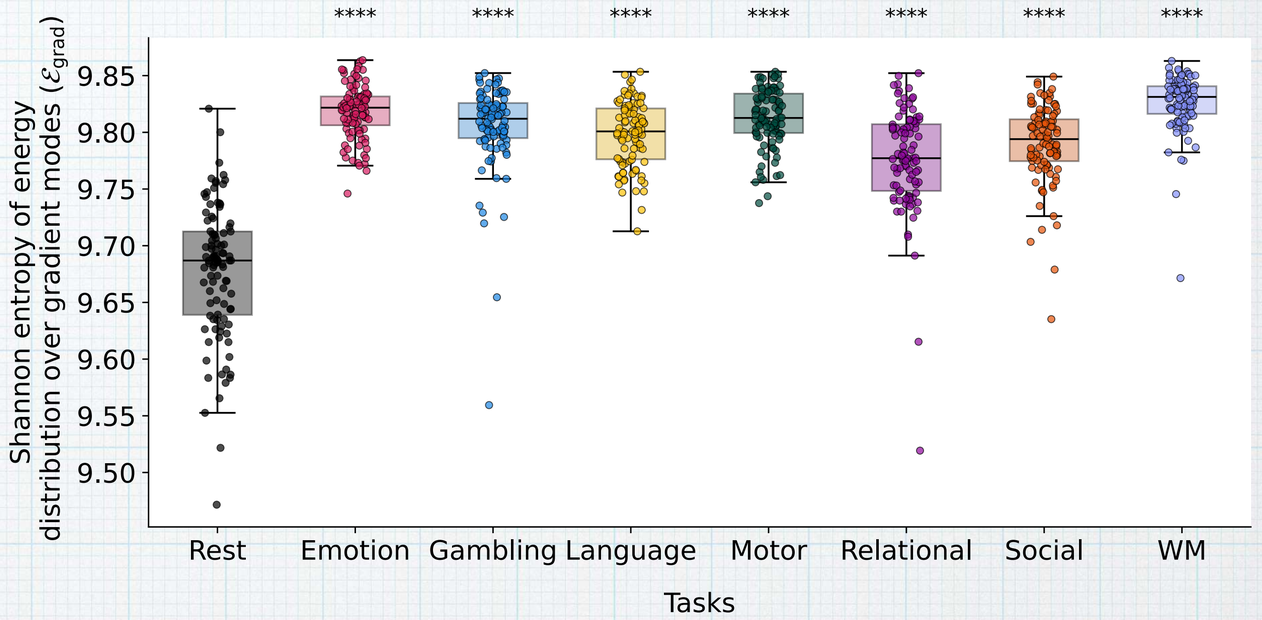
$$X(i,j) = \text{wrap}(\theta_j - \theta_i)$$

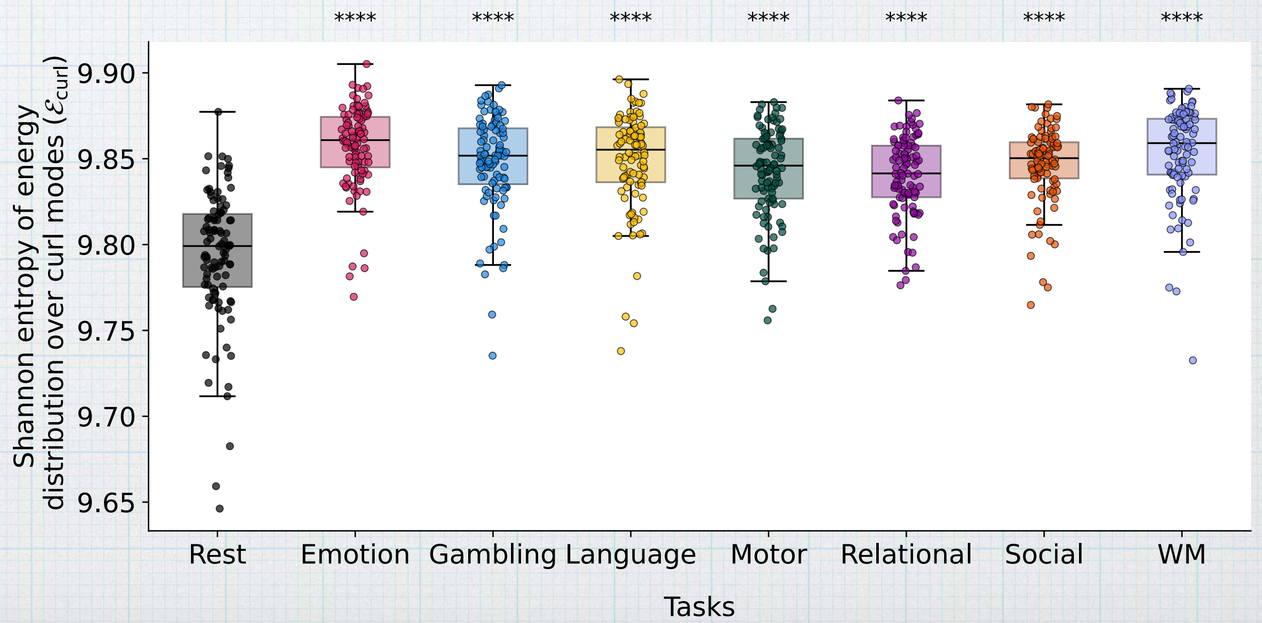
$$X(t) = \sum_{j=1}^{M_1} a_j(t)\xi_j + \sum_{j=1}^{M_2} b_j(t)\psi_j$$

$$E_U(k) = \langle |\Upsilon_U^k(t)|^2 \rangle_t,$$

$$P_U(k) = \frac{E_U(k)}{\sum_{l=1}^{Q} E_U(l)}.$$

$$\mathcal{E}_U = -\sum_{k=1}^{Q} P_U(k) \log\left(P_U(k)\right),$$





Quick recap

- Hodge theory on graphs defines vector calculus operations for discrete structures
- This allows for applications to dynamical systems like those from statistical physics
- Computations are easy; the mathematics is intricate but rigorous and principled

Thanks for listening

R. Nartallo-Kaluarachchi, M. Kringelbach, G. Deco, R. Lambiotte, A. Goriely (2026) Nonequilibrium physics of brain dynamics. *Physics Reports,* 1152, 1-43

R. Nartallo-Kaluarachchi, P. Expert, D. Beers, A. Strang, M. Kringelbach, R. Lambiotte, A. Goriely (2024) Decomposing force fields as flows on graphs reconstructed from stochastic trajectories. *Proceedings of the Third Learning on Graphs Conference*, PMLR 269 (Best Paper)

R. Nartallo-Kaluarachchi, R. Lambiotte, A. Goriely (2026) Coarse-graining nonequilibrium diffusions with Markov chains. Journal of Statistical Mechanics: Theory and Experiment, 033205

R. Nartallo-Kaluarachchi (2024) Cycles and potentials: from structure to stochastic dynamics. London Mathematical Society Newsletter (Sept, 24)

R.Nartallo-Kaluarachchi*, D. Beers*, E. Guzman, G. Deco, M. Kringelbach, R. Lambiotte, A. Goriely, M. Porter *coming soon*

Oxford Mathematics

UNIVERSITY OF OXFORD Mathematical Institute

UNIVERSITY OF CAMBRIDGE Faculty of Mathematics

Postdoc in DAMTP

with Mazi Jalaal

rdn23@cam.ac.uk

